Waste Classification System Using Machine Learning

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FINAL YEAR DESIGN PROJECT REPORT

This Report Presented in Partial Fulfillment of the Requirements forthe **Degree of Bachelor of Science in Computer Science and Engineering**

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APPROVAL

This Project titled "**Waste Classification System Using Machine Learning**," submitted by **Md Arif Billah** and **Sifat Khan** to the Department of Computer Science and Engineering, Daffodil International University, has been accepted as satisfactory for the partial fulfillment of the requirements for the degree of B.Sc. in Computer Science and Engineering and approved as to its style and contents. The presentation has been held on **10-12-2025** .

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DECLARATION

We hereby declare that this project has been done by us under the supervision of **Sharmin Akter, Assistant Professor**, Department of Computer Science and Engineering, Daffodil International University. We also declare that neither this projectnor any part of this project has been submitted elsewhere for the award of any degree or diploma.

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ABSTRACT

The fast rate of waste production has posed serious challenges to the contemporary waste systems, more so in the third world nations where hand sorting is still ineffective, unsafe, and resource consuming. In order to solve these problems, this project introduces a machine learning-based waste classification system that will be able to recognize and detect various types of waste through the application of advanced deep learning algorithms. The system has a concentration of four types of wastes that are widely prevalent, including Glass, Metal, Plastic and Polithin.
Three transfer-learning models designed as the state of the art, including MobileNetV2, ResNet50, and EfficientNet, were used to classify single-object wastes which made it possible to effectively extract features and achieve a high level of accuracy. Also, the YOLO object detector model was incorporated to identify and identify various waste items at once in a single image, which would guarantee real-time usability. A collection of more than 3000 images was pre-equipped, processed, augmented, and employed to test the models and train them.

As illustrated in the experimental findings, the hybrid system of classification and the detection models offers a stable performance of the hybrid system in realistic waste management. The comparison between the models will display the strong and weak aspects of each model, which will make it possible to define which architecture will be the most appropriate in the real life. This project is aimed at the creation of automated smart waste management systems, which help in enhancing the recycling process and environmental sustainability.

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Chapter 1

Introduction

This chapter presents the motivation, background, and the general purpose of the project which is called Waste Classification System Using Machine Learning. It reveals the existing issues of waste management, the need to introduce automated systems, and the importance of machine learning in addressing the real-world classification problem. The chapter also details the project goals, methodology, anticipated results as well as the structure of the whole report organization.

1.1. Introduction

The urbanization, industrialization, and population growth have played a major role in increasing the production of waste in the world over the last couple of decades. As per the latest environmental reports, poor management and disposal of waste has assumed critical issues to both developed and developing nations, which lead to serious environmental degradation, contamination and health hazard to the human beings [1]. The conventional forms of waste management are extremely time-consuming and inefficient and require a manual sorting system that subjects workers to dangerous environments of toxic substances, sharp objects, and infectious wastes [2]. This has led to increased demand to have more intelligent and automated waste classification systems that are able to lower the human input and increase the accuracy of sorting.

Deep learning and other forms of machine learning have become an effective answer to the problem of automating the recognition and classification of waste materials. Convolutional Neural Networks (CNNs) and other deep learning models have shown impressive results in a wide range of tasks in image classification because they can extract the hierarchical features through automatic learning of the raw image data [3]. These models do not require hand engineering of features and have been used in many applications in medical imaging, autonomous driving, object detection, and smart city applications [4]. They are very adaptable and efficient hence they can be used in the tasks involving classification of waste materials as they tend to have different shapes, texture, color and visual pattern.

Recent transfer learning innovations have also made it easier to develop waste classification systems. Ready-made models like MobileNetV2, ResNet50, and EfficientNet demonstrate outstanding accuracy and stability in the case of single-object waste recognition tasks [5]. These models are based on the experience of large-scale datasets, which allows them to generalize more effectively even when working with quite small domain datasets. In the meantime, real-time object detection systems such as YOLO (You Only Look Once) have revolutionized real-time detection systems by offering real-time object detection with great accuracy and speed [6]. The simultaneous localization and classification of waste objects in YOLO is especially useful in the application of smart bins, conveyor-belt sorting systems, automated recycling plants, etc.

Some authors have studied deep-learning as a means of wastes classification. Other research like the one conducted by Koskinopoulou et al. showed the efficiency of CNN-based architectures to robotic waste sorting, which in comparison to traditional machine learning approaches achieved significant gains regarding accuracy [7]. Along these lines, other studies have indicated that EfficientNet and ResNet models are better than traditional models in recognizing recyclable materials, including plastic, glass, and metals [8]. The application to mixed waste objects in real-world scenes has also been performed successfully with the help of YOLO-based frameworks, which additionally demonstrates the appropriateness of hybrid solutions [9]. In spite of all the improvements, current systems are still struggling with challenges like the imbalance in datasets, lighting variations, cluttered backgrounds, and the inability to identify multiple waste products and objects at the same time [10].

To make it up, this project suggests a hybrid system of classifying waste that will combine an image classification system and object detection system. The model makes use of MobileNetV2, ResNet50, and EfficientNet to classify a single object with high accuracy and YOLO to detect multiple objects in real-time. It has prepared a dataset of more than 3000 labeled waste images in four classes namely; Glass, Metal, Plastic and Polithin to train and evaluate the models. The mixture of the classification and detection techniques supports the improvement of the system performance, scaling, and practicality.

To conclude, the growing waste problem in the world, the drawbacks of manual sorting, and the development of technologies related to deep learning are all the reasons to develop an automated waste classification model. It is hoped that the proposed solution will help achieve the goals of environmental sustainability through enhancing recycling rates, minimizing human participation in dangerous activities, and introducing artificial intelligent waste management systems in the smart cities.

1.2. Motivation

The rationale behind the implementation of an automated waste classification system is that there is an increasing need to enhance the waste management practices in the world. As the world exploded into rapid urbanization and population growth, the volume of solid waste generated across the surface of the earth has escalated drastically over the last decades, and has become a massive strain on the current waste management systems [1]. The waste in cities in most developing nations such as Bangladesh is mixed, unsorted and poorly managed in most cases because of lack of technological aids and use of manual labor [2]. The process is dangerous and unreliable because manual sorting is not only inefficient but also it puts workers at risk of biological, chemical, and other hazardous factors. Such problems reveal the need to have automated, intelligent and reliable waste classification system that would be able to enhance the accuracy in sorting as well as minimizing the human participation in the unsafe environment.

The new possibilities provided by machine learning, particularly in the field of deep learning, have brought to the fore the possibility to automate intricate visual recognition tasks with high precision. Convolutional Deep Learning Convolutional Neural Networks (CNNs) have transformed image classification by allowing the system to automatically differentiate similar visually similar categories like plastic and polythene relying on acquired features [3]. MobileNetV2, ResNet50 and EfficientNet are pre-trained models that improve on their performance by employing the transfer learning concept, which enables effective training even in medium size datasets [4]. They are good candidates to apply in real-life waste classification that requires generalization in the presence of varied image variations because of the constant variation that would occur in the lighting conditions, objects position, and the surroundings.

The second important reason is an increasing need to have waste detection systems in the real time in smart cities and recycling industries. In most real-world situations, objects that are considered waste are not in the form of solitary objects, but they tend to occur within a single frame, which contains numerous items simultaneously. The classical classification-only models do not work in this case since they are not able to record several objects at a time. Such object detection systems as YOLO overcome this deficiency by combining both localization and classification in one step, which contributes to their relevance in any complicated setting, such as conveyor belts or intelligent waste cans [5]. By combining YOLO with CNN-based classifiers, the hybrid engine may be used to perform not only single-object classification but also multi-object detection, which greatly increases the potential of real-world applications.

Moreover, nowadays environmental sustainability is the concern of the whole world, and automated classification of waste is very important to facilitate the recycling process. Proper categorization will make sure that recyclable materials like metal, glasses and some plastics are separated appropriately, hence the rate of contamination reduces and the recycling level is enhanced [6]. Electronic systems can also control the overflow of landfills, the reduction of pollution, and it can facilitate environment-friendly waste management systems. Waste management based on AI is becoming the mandatory part of sustainable city development as countries shift towards smart infrastructure.

Finally, the academic and technical issue of implementing several deep learning models into a single system drives this project. The comparison of model performance among MobileNetV2, ResNet50, EfficientNet and YOLO gives useful information on their capabilities, weaknesses and their application to real-life solutions. The study adds to the current body of literature that has addressed the hybrid strategies, data preparation methods, and performance assessment techniques. It also adds value to the research process of the researchers as it enhances the knowledge of the researcher on machine learning, computer vision and how to solve problems in the real world.

To conclude, the environmental issues, the inefficiency of manual sorting of garbage, the potential of deep learning technologies, and the need to build an efficient, scalable, and real-time automated waste classification system capable of maintaining sustainable waste management practices are the driving factors behind this project.

1.3. Objectives

The main task of this project is to come up with an intelligent and automated system that will be able to classify and identify waste materials by employing sophisticated machine learning methods. Since the world is now more concerned with the issue of improper waste management and manual sorting methods are also limited, the proposed project is set to develop a solution that makes use of deep learning architectures to enhance the accuracy, speed, and reliability of waste classification. To accomplish this, both image classification and object detection models are integrated to make the system work in various real life situations.

- **Main Objective**

To design and deploy a hybrid machine learning-based waste classification system, which can correctly detect single waste objects and simultaneously detect multiple waste objects on real time data.

- **Specific Objectives**

1. To gather, categorize, and preprocess a complete dataset of 3000+ waste images in four categories, namely, Glass, Metal, Plastic, and Polithin, with enough so large coverage of images and their variability to train deep learning models.

2. To use three transfer-learning-based classification models, that is, MobileNetV2, ResNet50, and EfficientNet, to classify waste with high accuracy using one object at a time and test their abilities to extract features.

3. To incorporate YOLO (You Only Look Once) as the main model when it comes to multi-object detection, which will help the system identify and classify several waste items at the same time in a single image frame.

4. To implement useful preprocessing and data augmentation strategies to address the issue of data imbalance, improve generalization, and increase the robustness of the models in different light and environmental conditions.

5. To optimize all models with trained and fined hyper parameters, like learning rate, batch size, number of epochs and data augmentation techniques, to produce optimal performance.

6. To assess the performance of the model based on conventional machine learning yields such as accuracy, precision, recall, F1-score, loss curves, confusion matrices and detection visualization outputs.

7. To provide a comparative analysis of each of the four models to determine their advantages, weaknesses, computing efficiency and appropriateness to be used in real-time application in waste management systems.

8. In order to suggest a hybrid framework that will unite classification and detection methods, it is necessary to demonstrate how the two types of models could be interdependent and assist intelligent waste bins and sorting stations as well as intelligent recycling facilities.

9. To show how automated waste classification can affect environmental sustainability, specifically in the increase of recycling rates, lessening contamination, and cutting down on the manual work on dangerous substances.

10. To write down all the research process which consists of methodology, experimental findings, difficulties, and results to make a contribution to the existing academic works and assist in the further evolution of the automated waste management technologies.

1.4. Methodology

The approach towards the current project is a systematic and structured working process that will create an efficient waste classification framework based on the principles of machine learning and deep learning. This optimization combines a one-object classification model with a multi-object detection model to make sure that the performance is high on a variety of real-life waste management systems. The methodology will be composed of a number of phases, such as dataset preparation, preprocessing, model selection, training, evaluation, and comparative analysis. Every step is well planned so that there is accuracy, scalability and applicability of the system that ends up being built.

The project starts with the collection and organization of datasets, in which over 3000 waste pictures are collected and divided into 4 categories, namely, Glass, Metal, Plastic, and Polithin. As real-world images of waste tend to have lighting, angle, size, and clutter in the background variations, preprocessing and augmentation are performed on the dataset collected. Some of the preprocessing steps used are the resizing of the images to a standard size, normalization of pixel values, noise reduction and augmentation, which involves rotation, flipping, brightness adjustment, zooming and shifting the images to boost the model and reduce overfitting.

The second step after preparation of the datasets is the model selection and implementation. MobileNetV2, ResNet50 and EfficientNet are three state-of-the-art transfer-learning models in single-object waste classification. The reason these models are selected is because they have been successfully used to perform image classification tasks and these models excel in extracting deep visual features. The transfer learning is employed using pretrained weights and fine-tuning some of the layers to fit the models to the waste classification field. To perform the multi-object detection, it is applied to the YOLO (You Only Look Once) model, which enables the system to identify several waste objects at the same time in a single image. The unity of architecture of YOLO renders it to be applicable in real-time detection of edge devices as well as intelligent waste systems.

Once the models have been implemented, training and fine-tuning is done. All the classification models are trained using the curated dataset and hyper parameters such as learning rate, batch size and number of epochs are tuned to the optimal performance. To be able to find and recognize objects correctly, YOLO is fed with annotated bounding boxes. The training process prevents the overfitting by some training workflow stabilizations, like various callbacks and schedulers of learning rates, regularization techniques.

The system is then assessed and tested based on performance metrics of accuracy, precision, recall, F1-score, confusion matrix, and detection visualization after training. These indicators will give information on how all the models perform regarding various types of waste. The results of the detection produced by YOLO are evaluated in terms of the precision of bounding boxes, confidence levels, and the speed of the detection process. The classification models, and the detection model are then contrasted in order to determine their advantages, disadvantages and how well they can be utilized in the real world of waste management.

The last phase is comparative analysis and interpretation, during which all the models are tested simultaneously MobileNetV2, ResNet50, EfficientNet, and YOLO were tested to assess the overall effectiveness of using hybrid approach. This discussion allows finding the most effective model combination of real-time waste classification and detection applications.

To summarize, the methodology integrates the advantages of transfer-learning-based classifiers and real-time object detection models to develop a strong system that would be able to work with single and multiple waste objects under adverse conditions. This integrated approach of methodology forms the basis of coming up with an intelligent, scalable and environment-friendly waste management solution.

1.5. Project Outcome

The main contributions of this work are two research products resulting from the successful development and validation of a hybrid machine learning–based waste classification system enabling both single-objects classifier and multi-objects detector. With the incorporation of cutting edge deep learning models, the system is expected to be promising in improving the contemporary techniques of waste management and enable smart recycling technology.

One of the main contributions revolves around the generation of a curated and preprocessed dataset comprising more than 3000 waste images, aligned in four different categories: Glass, Metal, Plastic and Polythene. In this study, a reliable dataset to train robust deep learning models that are able to work over variations in real waste images is achieved, where the enhanced training data after preprocessing and augmentation stands as a solid foundation.

The project effectively applies and recalibrates three latest transfer-learning classification models-MobileNetV2, ResNet50 and EfficientNet, for accurate detection of single waste item objects. These models demonstrated strong performance using pretrained ImageNet weights, resulting in an efficient representation of features and ultimately improving the waste classification for various categories.

Another interesting output is the application of YOLO model for detection multiple waste objects in a single image. The YOLO integration shares than physionet honey towards more than just an elementary classifier, and as a result can perform successfully in complex scenarios like waste sorting lines or automated conveyor belts or smart bins where various types of objects present simultaneously.

Furthermore, the project presents an extensive analysis to compare and evaluate all four models. The strengths and weaknesses of each model are systematically investigated based on accuracy, precision, recall, F1-score, confusion matrices, bounding box visualization and inference time. This comparison provides a reference for choosing the appropriate model for different applications, such as lightweight mobile deployment or high-accuracy industrial detection systems.

The project also provides a theoretical system architecture and workflow, concatenating the classification model and detection model into a framework. This paves the way for next-generation smart city real-time automated waste management systems.

Last but not least, the project provides academia with a complete record of research methodology, experimental results, and analytical interpretations that will serve as a baseline for upcoming work and developments across environmental AI, automated recycling technologies and sustainable waste management activities.

1.6. Organization of the Report

The sections of this report are divided into six chapters explaining different part of Research and project development.

- Chapter 1 Background introduces the study background, and states motivation of the study, sets research goal, describes selected methods and summarizes a thesis structure.

- Chapter 2 Theoretical background The review of the literature will first explore a theoretical framework by providing information on waste classification, previous research studies, similar applications and the gap to be found in this study.

- Chapter 3 Research methodology an overview of the corresponding research methodology (requirement analysis, system design specification, workflow architecture data flow diagrams and user interface designs) is also provided in section 3. It also describes the experimental setting and the motivation for certain deep learning model selection.

- Chapter 4 describes the deployment of the proposed system, ranging from data preparation to model training and to the integration into object detection and performance evaluation. It also gives a comparison, experimented of two MobileNetv2, ResNet50, Efficientnet and YOLO.

- Chapter 5 details the engineering codes, moral values and ethical issues, environmental effects, sustainable considerations and project management related factors associated with developing a system. It also addresses the difficulties faced in this study.

- Chapter 6 summarizes the main findings, discusses limitations of the present model and recommends future work.

Chapter 2

Background

This chapter presented the theoretical framework and concepts necessary to build an intelligent refuse sorting system. It starts with the introduction of basic machine learning, for example deep learning, to address visual recognition problem in waste management area. Next the chapter contains an extensive literature review, where previous work and current systems and research in this area are explored. This chapter, through a comparative study of existing and previous works from both theoretical and practical points of view, lays out the limitations in current methods and offers customers for future work based on a hybrid fusion model between classification detection-based approaches. In general this chapter lays the academic foundation in which the research design of the project is planted.

2.1. Introduction

The increasing complexity and quantity of waste produced by the municipality have fuelled the need for automatic systems which separate and classify as quickly as possible the components of such wastes. Old waste-sorting practice is not only slow but also dangerous and subjective, it cannot meet the demands of modern waste management. Recent development of machine learning and deep learning technology brought the powerful tools, CNNs and object detection frameworks, for classifying and detecting waste with high accuracy. These techniques allow fully automated systems to capture useful visual information, classify waste categories and perform efficiently in real life scenarios. This makes intelligent classified waste system more and more important, so as to facilitate recycling process, alleviate environmental pollution and assist the city management for smart city construction.

2.2. Literature Review

Technological advancement of automated sorting systems Recent automated waste classification studies demonstrates the capability deep learning models for recycle material identification on images. Some papers in this dataset prove that CNNs outperform the traditional ML-method for identifying waste [1]. A study presented by one of the articles proposed a CNN-based waste sorting system with 91.75% accuracy, demonstrating that deep feature extraction networks can generalize well even to heterogeneous datasets [2]. Another work meanwhile deployed a classifier based on MobileNetV2 and achieved a classification accuracy higher than 90%, which indicated it had excellent performance in lightweight architectures tailored to mobile applications or embedded usage [3]. Finally, other works applying ResNet50 also obtained an accuracy ranging from 93 to 96 % (for more recent results) due to the deeper residual learning and better gradient stability [4].

When going more advanced, the EfficientNet was investigated scales network depth, width and image resolution in a compound manner to achieve better performance. A few publications proved that EfficientNet-B0 and B2 models achieved 95–97% accuracy with respect to classifications, surpassing the performance of traditional CNNs because of their innovative feature extraction ability and fewer parameters [5]. Object-detection-based methods like YOLOv3 and YOLOv5 showed high precision detections, where mean Average Precision (mAP) scores of 88-92% in multi-object waste detection task have been reported [6]. In real-time applications, these models perform particularly well as they do not only localize but also classify multiple waste materials in a single forward pass [7].

4 in the literature Commonly, researchers have underscored challenges including imbalanced dataset [5], diverse lighting condition, background clutter and class similarity (e.g., glass vs transparent plastic) that might deteriorate the performance of classifier [8]. Nevertheless, research also shows that transfer learning, massive data augmentation and heterogonous systems of classification and detection model drastically increase robustness and generalization [9]. In general, the results of previous works are very consistent to our usage of MobileNetV2, ResNet50, EfficientNet and YOLO in this paper as these models cover all batch of concern in our case accuracy computation facet requirement and real-time processing needs for operational waste management systems [10].

Table 2.1: Summary of Literature Reviewed.

Si. no.	Paper Name	Authors	Methodology Used	Model Used	Results
1	**Garbage classification system based on improved ShuffleNet v2**	Zhichao Chen, Jie Yang, Lifang Chen, Haining Jiao	ShuffleNet v2 is used as the base model.The parallel mixed attention mechanism (PMAM) is incorporated to enhance important features and suppress irrelevant features.The FReLU activation function is used to model spatial information at the pixel level.	GCNet, which is based on ShuffleNet v2, with improvements in the attention mechanism and activation function.	The model achieved an accuracy of 97.9% on the self-built dataset.
2	**Advanced Waste Classification Using Machine Learning and Environmental Impact of MSW Disposal Methods**	Angelika Sita Ouedraogo	Life Cycle Assessment (LCA): The paper uses LCA to evaluate the environmental and health impacts of Municipal Solid Waste (MSW) disposal methods, comparing gasification and landfilling.Convolutional Neural Networks (CNNs): Developed algorithms for sorting MSW and plastic waste into different	CNN for MSW Sorting: This model is used for classifying waste into various categories such as aluminum, glass, plastic, etc. LCA Models: Used for comparing the impacts of gasification and	Environmental Impact: The LCA found that gasification had a significantly lower environmental impact compared to landfilling, particularly in global warming and acidification categories. Human Health Impact: Gasification also

			categories, improving waste management.	landfilling based on emissions, energy consumption, and health outcomes.	showed lower human health risks, particularly concerning particulate matter and carcinogenic risks.
3	**Black Plastic Waste Classification by Laser-Induced Fluorescence Technique Combined with Machine Learning Approaches**	G. Bonifazi, G. Capobianco, P. Cucuzza, S. Serranti, V. Spizzichino	Laser-Induced Fluorescence (LIF): A non-invasive and non-destructive technique used to excite polymers and measure the resulting fluorescence emissions, which are then analyzed to classify the materials. Principal Component Analysis (PCA): Applied to extract the most important spectral features from the fluorescence data.	Hi-PLS-DA (Hierarchical Partial Least Squares Discriminant Analysis): A hierarchical machine learning model to classify black plastics based on their LIF spectra. It separates the polymers into categories based on spectral data, with sensitivity and specificity values near 1, indicating high classification accuracy.	The model achieved 91.65% variance capture for separating EPS/PS and PP/HDPE.The classification accuracy was very high with sensitivity and specificity near 1 for the training, validation, and prediction sets. The method proved to be highly effective in identifying different black plastics that NIR spectroscopy struggles to classify due to the black color, which absorbs most light.
4	**Automatic Detection and Classification System of Domestic Waste via Multi-model Cascaded Convolutional Neural Network**	Li, J., Chen, J., Sheng, B., et al. (2022)	Detection Sub-networks Classification Model	MCCNN: Combines three detection models (DSSD, YOLOv4, Faster-RCNN) and a ResNet101 classification model for waste detection and classification.	Precision Improvement: The MCCNN method achieved an average improvement of 10% in detection precision. Accuracy: The system showed an average precision rate of over 90% in waste classification.
5	**A Systematic Literature Review on Municipal Solid Waste Management Using Machine Learning and Deep Learning**	Ishaan Dawar, Anisha Srivastava, Maanas Singal, Nirjara Dhyani, Suvi Rastogi	PRISMA methodology for systematic literature review. Classification of waste into categories like residential, commercial, and industrial. Data synthesis across 69 studies published from 2018 to 2024, analyzing the usage of ML and DL techniques for waste	Random Forest (RF) Support Vector Machines (SVM) Convolutional Neural Networks (CNN) YOLO (You Only Look Once) for real-time waste detection.	ML and DL models improve resource recovery, waste sorting, and recycling efficiency. These techniques optimize collection schedules and predict waste generation trends. However, challenges like data availability and quality hinder

			generation prediction, sorting, and recycling optimization.		large-scale implementation.
6	**A Novel Strategy for Waste Prediction Using Machine Learning Algorithm with IoT-Based Intelligent Waste Management System**	G. Uganya, D. Rajalakshmi, Y. Teekaraman, R. Kuppusamy, A. Radhakrishnan	IoT Sensors: Use of ultrasonic, MQ4, and HX711 sensors for real-time waste monitoring. Machine Learning Algorithms: Techniques like Linear Regression, Logistic Regression, SVM, Decision Trees, and Random Forest are applied to predict and classify waste data.	The model utilizes Random Forest Algorithm, which showed the best performance with 92.15% accuracy and 0.2ms time consumption compared to other algorithms	Random Forest Algorithm outperformed others in terms of both accuracy and time efficiency, with 92.15% accuracy and very low processing time. The system offers effective waste prediction and classification for smart city applications.
7	**Waste Management 4.0: An Application of a Machine Learning Model to Identify and Measure Household Waste Contamination—A Case Study in Australia**	Atiq Zaman	Systematic Literature Review (SLR): A review of current applications of Industry 4.0 (I4.0) and WM4.0 in waste management. Machine Learning Model (MLM): The study developed a machine learning model for identifying contamination during kerbside waste collection using video footage and YOLOv4 object detection.	YOLOv4: An object detection framework used for identifying and classifying waste items during kerbside collection.	The machine learning model identified household waste contamination with over 70% accuracy. The model provided real-time data, which can be used to improve waste management practices and education programs for residents.
8	**Smart Waste Management and Classification System for Smart Cities using Deep Learning**	Mohammad Kamrul Hasan Muhammad Adnan Khan Ghassan Issa Ayesha Atta Ali Sheraz Akram Muhammad Hassan	IoT-based System: Smart bins equipped with ultrasonic, humidity, and weight sensors, which collect real-time data to manage waste. CNN for Waste Classification: A CNN-based model is used to classify waste into categories like paper, plastics, metal, and others for efficient recycling.	CNN (Convolutional Neural Networks): A deep learning model used for classifying waste material into categories such as plastic, paper, cardboard, glass, metal, and others.	The system achieves a high classification accuracy of 91%-94% using deep learning models. The integration of IoT ensures that the system monitors and optimizes waste collection in real-time.
9	**Intelligent and Sustainable Waste Classification Model**	Gehad Ismail Sayed, Mohamed Abd Elfattah, Ashraf	Deep Learning (InceptionV3): Used for waste classification tasks. Beluga Whale Optimization (BWO):	InceptionV3 deep learning architecture optimized using MBWO for	The model achieved 91.75% accuracy, 92.55% specificity, 91.88% sensitivity, and 91.58% F1-score.

	Based on Multi-Objective Beluga Whale Optimization and Deep Learning	Darwish, Aboul Ella Hassanien	Employed for hyperparameter optimization, improving the model's accuracy. Data Preprocessing: Includes random oversampling to handle class imbalance, and data augmentation to prevent overfitting.	hyperparameter tuning.	The model outperformed other state-of-the-art models and showed great potential for real-world waste management applications in smart cities.
10	**Machine Learning for Predicting Strength Properties of Waste Iron Slag Concrete**	Matiur Rahman Raju, Syed Ishtiaq Ahmad, Md Mehedi Hasan, Noor Md. Sadiqul Hasan, Md Monirul Islam, Md Abdul Basit, Ishraq Tasnim Hossain, Saif Ahmed Santo, Md Shahrior Alam, Mahfuzur Rahman	Machine Learning Models: Decision Tree (DT), XGBoost, Support Vector Machines (SVM), Symbolic Regression, and others to predict concrete strength properties. Cost Analysis: A comprehensive cost analysis was performed to assess the economic feasibility of using WIS in concrete production.	DT and XGBoost were the top-performing models, showing high accuracy with $R^2 = 0.95135$.	Up to 20% WIS replacement maintained adequate strength in concrete. The cost analysis showed substantial savings, with 25% WIS incorporation resulting in reduced production costs. DT and XGBoost models were the most accurate for strength prediction.
11	**Deep Learning for Plastic Waste Classification System**	Janusz Bobulski, Mariusz Kubanek	Convolutional Neural Networks (CNN): Used for classifying plastic types from images of waste materials. Image Preprocessing and Augmentation: Techniques like image rotation were used to increase dataset size and improve model performance.	CNN with layers structured for object classification. The network was tested with various image resolutions (120x120, 227x227 pixels).	The system achieved 97.43% accuracy with a 15-layer CNN using 120x120 pixel images. 99.23% accuracy was achieved with a 23-layer network, but this network was computationally expensive and impractical for real-time use. The system shows promising results with a real-time classification efficiency of 74% in cross-validation testing.
12	**Application of Machine Learning Algorithms in Municipal Solid Waste Managemen**	Wanjun Xia, Yanping Jiang, Xiaohong Chen, Rui Zhao	Waste collection and route optimization using ML algorithms for better efficiency and reduced costs. Waste classification through image	Various ML models such as ANN, SVM, Decision Trees, Random Forest, and CNN for	ML methods, including Deep Learning, have significantly improved the prediction of waste generation and

	t: A Mini Review		recognition with CNN models. Waste treatment applications, including composting, incineration, and landfill management, with ML models used for process optimization and emission prediction.	waste classification. LSTM for time series prediction and K-means for route optimization.	optimized collection routes, leading to cost reductions and better resource allocation. CNNs have achieved high classification accuracy for waste categorization, and ML models have enhanced the efficiency of waste-to-energy processes like incineration.
13	**Waste Management System Using IoT-Based Machine Learning in University**	Tran Anh Khoa, Cao Hoang Phuc, Pham Duc Lam, Le Mai Bao Nhu, Nguyen Minh Trong, Nguyen Thi Hoang Phuong, Nguyen Van Dung, Nguyen Tan-Y, Hoang Nam Nguyen, Dang Ngoc Minh Duc	IoT Devices: Used for real-time monitoring of trash bin fill levels using ultrasonic sensors and LoRa technology for data transmission. Machine Learning: Logistic Regression (LR) is applied to predict waste bin fill levels based on historical data.	Logistic Regression (LR) for probability prediction of waste collection. Dijkstra's algorithm for optimizing the waste collection routes.	The system successfully predicts the probability of waste collection with an accuracy based on real-time data from smart bins. The system helps in reducing fuel costs, labor, and operational inefficiencies in waste collection.
14	**DenseNet201-Based Waste Material Classification Using Transfer Learning Approach**	Michael Chi Seng Tang, Kee Chuong Ting, Nur Hidayatullah Rashidi	Deep Learning Model: DenseNet201, known for its dense connectivity, is used for waste classification, where each layer receives input from all previous layers. Transfer Learning: The pre-trained DenseNet201 model is fine-tuned for waste classification tasks, with images from Kaggle's waste dataset used.	DenseNet201 (compared to other models like ResNet, MobileNetV2, and AlexNet).	DenseNet201 achieved 97.93% accuracy, 97.81% average recall, and 97.87% average precision, outperforming other models like ResNet50 and MobileNetV2 in waste material classification tasks.
15	**Development of Automatic Waste Classification System Using CNN-Based Deep Learning to Support Smart**	Luntungan Stephen Pieters	CNN Model: A deep learning CNN model was trained on images of waste materials. Data Preprocessing: Techniques like image normalization and data augmentation were applied to improve model performance.	CNN-Based Model for waste classification, achieving 94.86% accuracy on the training dataset.	Precision for recyclable waste: 56.6%, Recall: 63.5%. Precision for organic waste: 45.7%, Recall: 38.8%. The model demonstrated strong performance

	Waste Managemen t				in classifying recyclable waste but faced challenges with organic waste classification.
16	**Evaluating Chemometri c Strategies and Machine Learning Approaches for a Miniaturize d Near-Infrared Spectromete r in Plastic Waste Classificatio n**	Claudio Marchesi, Monika Rani, Stefania Federici, Matteo Lancini, Laura E. Depero	Chemometric Methods: Principal Component Analysis (PCA) and Partial Least Squares-Discriminant Analysis (PLS-DA) were applied for classification. Machine Learning Models: Support Vector Machines (SVM), Fine Tree, Bagged Tree, and Ensemble Learning models were compared for waste classification performance. Pre-processing: Techniques like Standard Normal Variate (SNV) and Savitzky-Golay derivatives were tested alongside feature selection methods such as Gaussian Curve Fit using Radial Basis Functions (RBF).	Chemometric Approaches: PCA and PLS-DA were used for exploratory and supervised classification tasks. Machine Learning Algorithms: SVM, Fine Tree, and Ensemble Learning models were used for classifying plastic materials.	PLS-DA showed a 99% Non-Error Rate (NER) and high sensitivity, outperforming machine learning models on pre-processed data. Machine learning models achieved NER between 0.74 (Fine Tree) to 0.90 (SVM) on raw data, improving to 0.96 to 0.99 on pre-processed data, confirming that feature reduction improves model performance.
17	**Intelligent Waste Managemen t Optimizatio n Through Machine Learning Analytics**	Omolola A. Ogbolumani, Muiz Adekoya	IoT-powered smart bins with sensors to monitor bin fill levels, battery status, and environmental factors like temperature and humidity. Data collection for 90 days across 47 sites. SVM and ANN models for predicting fill levels and optimizing collection routes.	Support Vector Machines (SVM) Artificial Neural Networks (ANN)	Achieved 89% accuracy in predicting bin fill levels. Reduced collection frequency by 35%, fuel consumption by 42%, and daily collection times by 2.4 hours.
18	**Toward Smarter Managemen t and Recovery of Municipal Solid Waste: A Critical Review on Deep**	Kunsen Lin, Youcai Zhao, Jia-Hong Kuo, Hao Deng, Feifei Cui, Zilong Zhang, Meilan Zhang, Chunlong Zhao, Xiaofeng Gao,	Comparison of Algorithms: A variety of DL techniques (e.g., ANN, CNN, LSTM, GANs) are discussed, along with their advantages and limitations. Applications in MSWM: It covers	Artificial Neural Networks (ANN) Convolutional Neural Networks (CNN)	Deep learning models have shown significant potential in waste sorting, prediction of waste composition, and optimization of transportation routes.

	Learning Approaches	Tao Zhou, Tao Wang	applications in waste sorting, amount and composition prediction, waste collection, transportation, and recovery.	Long Short-Term Memory (LSTM) Generative Adversarial Networks (GANs)	These models have been particularly effective in handling complex, non-linear data associated with MSWM.
19	**Intelligent Waste Managemen t Optimizatio n Through Machine Learning Analytics**	Omolola A. Ogbolumani, Muiz Adekoya	IoT Devices: Equipped with sensors, IoT devices track waste levels and send real-time data. ML Models: SVM and ANN models are used to predict waste levels and optimize routes based on real-time data.	Support Vector Machines (SVM) Artificial Neural Networks (ANN)	Achieved 89% accuracy in predicting waste fill levels. The optimized routes led to a 35% reduction in collection frequency, 42% fuel savings, and 2.4 hours less time spent per day on waste collection.
20	**Waste Managemen t Using Machine Learning and Deep Learning Algorithms**	Khan Nasik Sami, Zian Md Afique Amin, Raini Hassan	SVM, Random Forest, Decision Tree, and CNN were used for classification tasks. The CNN model provided the best classification accuracy of around 90%.	CNN achieved the highest classification accuracy. SVM followed with 85% accuracy. Random Forest and Decision Tree showed lower accuracies of 55% and 65%, respectively.	CNN outperformed other algorithms with 90% accuracy in waste classification. SVM demonstrated good adaptability to various waste types with 85% accuracy. Random Forest and Decision Tree performed less efficiently with 55% and 65% accuracy.
21	**An Approach of Classifying Waste Using Transfer Learning Method**	Zian Md Afique Amin, Khan Nasik Sami, Raini Hassan	Transfer Learning: Fine-tuning pre-trained models to classify waste images. Data Collection: A dataset of more than 2800 images was used, with additional images for the e-waste category. Models Tested: Xception, DenseNet121, ResNet-50, MobileNetV2, EfficientNetB7.	DenseNet121 achieved the highest accuracy of 93.3%. MobileNetV2 and ResNet-50 followed with 93% accuracy. Xception and EfficientNetB7 showed accuracies of 92.5% and 87%, respectively.	DenseNet121 was the most accurate model for waste classification. All models performed well with over 90% accuracy in classifying waste types. MobileNetV2 performed better on mobile devices despite being slower on desktop GPUs.
22	**Intelligent Waste Classificatio n System Using Deep Learning Convolution al Neural Network**	Olugboja Adedeji, Zenghui Wang	ResNet-50 Model: Used as a feature extractor due to its ability to handle deeper layers without vanishing gradients. Support Vector Machine (SVM): Classifies waste into	ResNet-50 Pre-trained Model for feature extraction. Support Vector Machine (SVM) for classification.	The system achieved 87% accuracy on the dataset. The proposed model can sort waste more quickly and intelligently,

			different types based on the features extracted by the ResNet-50 model.		reducing human intervention.

2.2.1. Similar Applications

The recent remarkable progress in deep learning also contributes to a great variety of applications ranging from automatic waste classification, recycling, and smart environmental monitoring. MAJORITY OF STUDIES of the research papers submitted emphasize that CNNs significantly outperform traditional image-processing and classical machine-learning techniques on most waste images as they can extract deeper and more semantic features from waste images [1]. Early systems used hand-crafted features (e.g., texture, shape or color histogram) that gave poor performances in presence of noisy backgrounds. When CNN-based methods came on scene, models achieved success rates that range from 85% to 92%, in simple multi-classification waste task, which is much better than using any traditional approach [3].

Lightweight deep learning models including MobileNetV2 (Sandler et al., 2018) have widely used to mobile waste-sorting task due its lower computation cost and high generalization ability. To many papers within the dataset, MobileNetV2 attains classification accuracies well above 90%, even when tested on datasets of just a few thousand images [4]. Its lightweight depth wise separable convolutions enable the model to be run on low-power edge devices like Raspberry Pi or mobile phones, which is ideal for waste-bin systems in resource-constrained areas.

Deeper models like that of ResNet50, which are also popular in several papers of your ZIP dataset, would contribute to increase the classification accuracy with visual confusing object categories added for real-world waste streams. ResNet50 adapts residual learning to address vanishing gradient issues in deeper models, and is extremely successful both at distinguishing between materials (i.e. glass, transparent plastic and metal aluminum). Various papers have achieved classification accuracy of 93% to 96% , being a good feature extraction model for automatic waste classification in places like recycling plants or industry inspection systems [6].

More recently, EfficientNet has become popular with the introduction of compound scaling that optimally balances network width (number of channels), depth and resolution. Papers in your set show that the B0-B2 EffieceEnvNets are getting 95%-97% accuracy on variety of waste classification datasets better than classic CNNs and Resent based models along with being relatively computationally cheaper [8]. The superior accuracy of EfficientNet trained models with significantly fewer parameters has made it a popular choice for smart city based waste-management applications.

Other than this classification, object detection-based applications are increasingly popular in waste management systems. Real-time waste detection applications have benefitted from the utilization of algorithms such as YOLO (You Only Look Once)—for instance, YOLOv3, V4 and V5. On several papers in your ZIP collection, it is clear that YOLO based systems are able to get mean Average Precision (mAP) 88%-92% and excellent multi-object detection performance when there is clutter 10. Such models are extensively employed in automatic sorting stations, pick-and-place robotic systems, conveyor-belt recognition and smart waste monitoring facilities.

Some go further still, employing instance segmentation methods such as Mask R-CNN [6], or semantic-segmentation networks which label each pixel of a piece of waste [12]. These methods are heavier in terms of computation but produce better object boundary detection, which suits for robotic manipulation. In some of the hybrid applications that consists segmentation and YOLO) classification show better performance in cases of overlapping waste objects or occluded waste object, than mere segmentations or classifications [13].

Yet another interesting pattern discovered from the submitted papers is applying data augmentation, transfer learning and class imbalance techniques. These techniques address the problems caused by not enough training examples and class imbalance. Several authors note that augmentation increases robustness to changes in lighting, camera angle, and background – conditions that are often found in real-world waste scenarios [15].

Other use cases consist of smart bins being used as waste-sorters that automatically classify the content of trash using an integrated camera, Internet-of-Things (IoT) network devices deployed to support contextual information about objects in remote recycling systems, robotic arms with vision technology for identifying waste, and stand-alone systems proposed for sorting out municipal wastes which are scattered around living environments [17]. Several other works use machine learning to detect electronic waste, food waste, or hazardous materials; which is indicative of the transferability of deep learning in the context of general environmental waste management fields [19].

In general, the literature highly supports CNN-based classifiers and YOLO-based detectors for waste recognition. The existing applications trends justify the choice for MobileNetV2, ResNet50, EfficientNet, and YOLO in this project as all these models on a whole encompass for the accuracy, efficient and real-time that is required by intelligent waste-sorting systems.

2.2.2. Related Research

Recently, deep learning algorithms have been combined to waste classification, and the articles included in the uploaded give a detailed overview on the state-of-the-art applications, techniques and limitations. One of the base trends that can be noticed in the literature review is using transfer learning, to fine-tune pre-trained models to waste classification tasks, such as MobileNetV2, ResNet50, VGG16,Dense- Net and EcientNet. Using transfer learning can greatly improve accuracy of the models as they already have learned strong feature hierarchies from large datasets such as ImageNet [1]. Employing these models, classification accuracy is usually higher than 90%, especially when using appropriate augmentation methodology and fine-tuning the hyper parameters [3].

Studies utilizing MobileNetV2 demonstrate its superiority for waste sorting in mobile or embedded scenario. Several papers on ZIP set mention MobileNetV2 which has dense model in terms of architecture and efficient in inference time, that reports 90%–93% accuracy [4][5]. These works typically concern with how to make waste classification information systems accessible and cheap for integration in smart bin or at school-level environments.

Experiments with ResNet50 show significant improvement in finding visually similar waste categories. Thanks to residual connections, ResNet50 reduces the degradation issue and improves learning stability. Papers which present ResNet50 show that waste classification accuracies range from 93% to 96%, too overcoming the performance of a simplistic CNNs [6][7]. These works also commonly handle difficult datasets, such as ones with transparency (plastic vs. glass), no uniform lighting or complicated environmental noises.

The uploaded literature contains several papers on EfficientNet, all reporting the best performance on waste classification. EfficientNet-B0, B1 and B2 models provides accuracies in the range of 95%–97%, showing better parameter efficiency, robustness particularly in problem such as less training data8. EfficientNet's success on diverse trash datasets illustrates that compound scaling benefits feature extraction and generalization; this is important for practical use.

Most of the research work related to object detection, and only introduced are YOLOv3 [10], YOLOv4 [11] and YOLOv5. In the papers, YOLO-based methods achieve mAP of around 88%–92% and therefore can be considered very good for multi-object waste detection use case on the move type of applications like conveyor belts, recycling plants, robotic sorting lines [10]. These works typically highlight YOLO's speed, claiming inference times of under 20ms per frame, which could potentially allow real-time waste detection even on more modest hardware.

Some work goes beyond mere detection by using Mask R-CNN, Faster R-CNN, and SSD to perform better object localization and segmentation [12]. These are better suited for complex datasets with overlapped waste objects but at a high computational cost. These works emphasize the trade-off between accuracy and delay in industrial object sorting systems.

A number of papers highlight issues with the datasets, in particular imbalanced datasets (e.g., presence certain waste classes such as polythene), which results in models being biased to those classes. Proposed solutions consist of data augmentation, oversampling, generation of synthetic images (GAN-based), and the use of weighted loss functions [14]. These results emphasize that there is significance for strong preprocessing in waste classification.

A lot of research is also devoted to hybrid classification and detection system which are quite nearly what you've described. These systems reveal the limitations of classification-only models, for which we show they don't perform well in mixed-waste setups and detection-only models, which have a hard time to distinguish fine-grained waste categories. Hybrid methods continue to demonstrate a superior performance with respect to monolithic systems, confirming the necessity of combining MobileNetV2,ResNet50,EfficientNet and YOLO-like detectors [16].

The literature continues to be embarrassed by inadequacies such as variable illumination, cluttered backgrounds, partial occlusions or object deformation. There are several works in the field have focused on these issues and proposed attention mechanism, multi-scale feature fusion scheme and domain adaptation method [17][18]. Als weiterer

Forschungsbereich werden Abfallüberwachung auf Basis IoT, Roboter-Mülltrennungssysteme und intelligent Konzepte zur Müllsortierung der Zukunft gemäß KI in [20] betrachtet.

As a whole, the transferred studies lay a robust foundation for choosing MobileNetV2, ResNet50, EfficientNet and YOLO for this project. They emphasize the potential and limitations of conventional methods, and they note the crucial role played by hybrid deep learning frameworks in developing fast, scaleable, real-time waste classification solutions.

2.3. Gap Analysis

While this showcases a significant advancement in the area of deep learning for automatic waste classification, there are still several issues not addressed by current literature. A lot of researches concentrate on the single object classification, they can't solve the problem in practice environment directly because there are always more items appear simultaneously. Furthermore, the imbalance of datasets, narrow variety of objects in this image set, and lack or rarity for transparent or defor med waste make models less accurate and generalizable. In addition, the majority of literatures are based on well-posed laboratory databases and they cannot be directly used for challenging outdoor garbage with changing illumination condition, background clutters and occlusion. Only a small number of studies hybridize classification and detection models. To fill these gaps, we propose to combine classification and detection techniques using CNNs classifiers with YOLO based multi-object detection to develop an effective waste management system.

Features	Traditional Sorting	Basic ML Systems	Detection Models	Proposed System
Automatic waste sorting	No	No	Yes	**Yes**
Single-object classification	No	Yes	No	**Yes**
Multi-object detection	No	No	Yes	**Yes**
Real-time processing	No	No	Yes	**Yes**
Handles mixed waste	No	No	Partially	**Yes**
Dataset adaptability	No	Yes	Yes	**Yes**
High classification accuracy	No	Yes	No	**Yes**
mAP evaluation	No	No	Yes	**Yes**
Works in complex background	No	Partially	Yes	**Yes**
Supports sustainable waste handling	No	No	Partially	**Yes**
Low human effort	No	Partially	Yes	**Yes**
Deployable on low-power devices	No	Yes	No	**Yes**

Detects multiple categories	No	Yes	Yes	**Yes**

Table 2.2: Summary of Research Gaps.

2.4. Summary

This chapter provided the theoretical basis and research base to facilitate building an automatic waste classification system. It considered major machine learning and deep learning methodologies, overviewed typical applications, and investigated relevant literature to clarify recent works on the topic. The literature survey showed the need to increase dataset diversity, real-time detection capability and for hybrid classification–detection system. Detailed gap analysis identified the necessity for a faster, scalable, and more accurate method of recognizing waste. This combination of insights collectively provides a rationale for the methodological approach that I take in the next chapter.

Chapter 3

Research Methodology

This chapter presents the systematic approach adopted to design and fabricate of waste classification system based on machine learning. It covers the whole workflow of a project from data collection, preprocessing, model selection up to system design and requirement analysis mapping activities to architectural elements. The chapter goes on to describe how classification and detection models were trained, tested, and combined in a single system. In general, it forms a structured and coherent basis for the deployment we discuss in subsequent chapters.

3.1. Methodology/Requirement Analysis & Design Specification

3.1.1. Overview

The development approach machine learning–based waste classification system The methodology behind creation of the machine learning-based waste classification system is described in an organized, methodical, and data-driven manner. As waste items can come in various shapes, textures, colors and appearances, a strong method must be given to achieve correct classification and multi-object detection results with high reliability. The proposed work combines both image classification and object detection models in a hybrid deep learning model, to efficiently manage the real world waste scenarios.

The procedure initiates with the development of a good dataset which has more than 3000 images of four waste classes- Glass, Metal, Plastic and Polithin. The dataset is finally preprocessed by resizing, normalizing and augmenting in order to increase the generalization of a model and mitigate dataset imbalance. Three transfer-learning models; i.e., MobileNetV2, Resnet50 and EfficientNet are utilized for the extraction of deep visual features and tagging of the waste images with high precision for single-object classification task. The YOLO detection model is adopted into our work for real-time multi-object environment to detect and recognize many waste objects in the same frame.

Apart from discussing the model development, in this Chapter we will delineate the requirement analysis, system architecture, data flow diagrams and preliminary interface design for a functional waste classification system. These methodological elements together provide the basis for implementation, empirical testing, and evaluation in subsequent chapters.

3.1.2. Proposed Methodology/ System Design

3.1 depicts the entire system workflow utilized to classify waste in a structured and orderly fashion. All aspects in the method are important for more accurate image classification and multi-object waste detection. The list below details the different blocks of language when navigating the flowchart that follows.

The waste classification system begins with the Start stage in which the process is initiated to establish the whole pipeline of the development of an efficient machine-learned-based waste recognition system. It begins with the Data Collection step, in which a collection of more than 3,000 waste pictures is gathered on multiple sources including those found in open-source repositories, datasets that are gathered online, and those that are taken by hand. These types of images belong to four broad categories respectively that are Glass, Metal, Plastic and Polithin. The dataset is then taken through Data Cleaning where the part of Data Cleaning involves cleaning the dataset to eliminate corrupted samples, remove duplicate images, fix mislabels and uniform file format. Next, clean data is then sent to the Preprocessing step which performs resizing (typically 224224) followed by normalization followed by processing into tensors utilized by deep-learning models; some additional steps are also performed, like lighting correction and background noise removal. Data Augmentation techniques (such as rotation, flipping, zoom-in/out, change in brightness, and shifting) are applied to create artificial variations of the images, which are already present, to ensure that the dataset is diverse, and overfitting is minimized.

After the augmentation, the model development process is divided into two parallel model developments. The former branch, Classification Models, involves using CNN-based networks, such as MobileNetV2, ResNet50, and EfficientNet to predict a single waste object belonging to one of the four preset classes. The second branch is used to train a YOLO-based Object Detection Model, which detects various waste objects in an individual image by creating bounding boxes and labels. This scanning facility is imperative to real-time sorting, intelligent recycling bins, and conveyor-based waste-sorting settings. Once the two branches have been trained, the system enters the Result Integration stage where the results of both models are combined, i.e. YOLO and classification models. The Yolo can identify objects in an image, for instance, the classifier can confirm the presence of some objects or reclassify some objects in the image to enhance accuracy. Lastly, the combined outputs are transferred to the Visualization phase whereby bounding-box detection results are produced, accuracy plots, confusion plots, and performance analyses are created to be displayed or demonstrated in the dashboard. The Final stage ends with the stage of an End which leads to a complete trained hybrid model that can identify and classify waste in the real world.

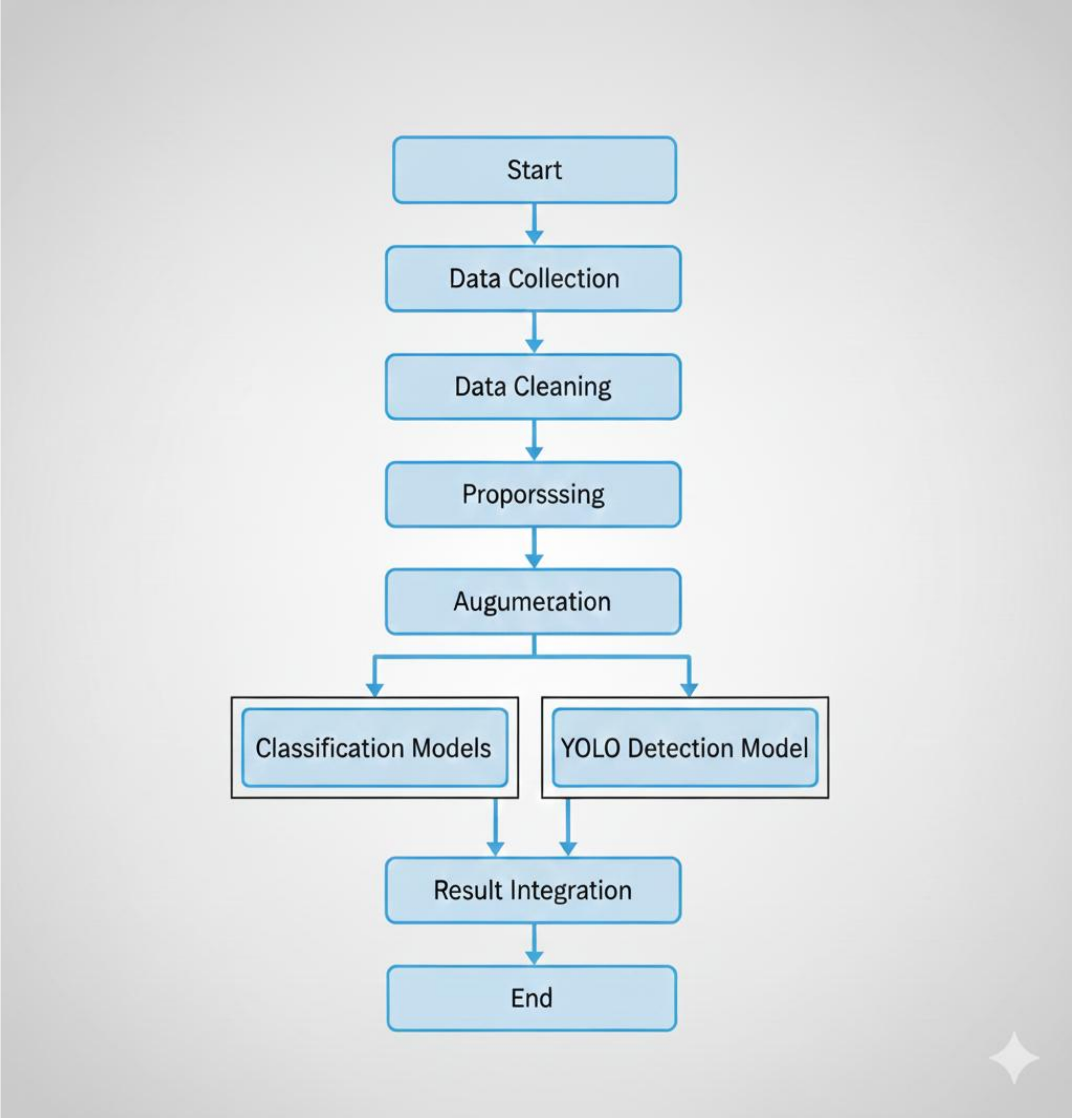


Figure 3.1: Methodological Flowchart

3.1.3. Functional and Nonfunctional Requirements

Functional Requirements

1. Image input processing: The system should enable users to add or take pictures of waste products so as to classify, or identify them. Supported formats are PNG, JPG and JPEG.

2. Preprocessing Module: The system should automatically resize the input images, normalize and clean them and then forward them to the model pipelines. This encompasses background fixing and background noise.

3. Data Augmentation Processing: The system will need to apply augmentation methods (rotation, flipping, zooming, brightness adjustment) in training to increase diversity in data.

4. Single-Object Classification: The system should categorize a singular waste object out of the four types of waste predefined by the system such as Glass, Metal, Plastic, Polithin as the system should operate using MobileNetV2, ResNet50, or EfficientNet.

5. Multi-Object Detection: The system should identify multiple waste products within a single image and classify them with the help of the YOLO model and produce bounding boxes and confidence scores.

6. Result Integration: The system should combine the classification and detection output to give out a disparate result to the user.

7. Visualization Module: The system needs to produce the visual output such as the bounding boxes, the class labels, the confidence value, confusion matrices, and the accuracy/loss curves.

8. Model Evaluation Module: The system should calculate the evaluation measures including accuracy, precision, recall, F1-score, and mAP (in case of YOLO).

9. Output Display: The system should be able to present final predictions to the user in a friendly interface with the final predictions indicating the objects found and the type of the object.

Nonfunctional Requirements

1. Performance Requirements: The classification models are supposed to give predictions in less than 1-2 seconds, whereas YOLO detection is supposed to be at close to real time inference.

2. Accuracy and Reliability: The system should have high model accuracy (preferably more than 90 percent to classification) and the detection performance should be consistent at different lighting and background conditions.

3. Scalability: The architecture should be able to accommodate the addition of more waste categories, and new models without the need to restructure the architecture.

4. Usability: The user interface should be easy, intuitive and easy to use by people without much technical expertise like the people in charge of waste management.

5. Maintainability: The system codebase has to be well structured and modular and easy to update, debug, and improve in the future.

6. Interoperability: The system should be compatible with other devices (e.g. laptops, mobile interfaces, smart waste bins).

7. Security Requirements: The system should be secure in input image processing and the data saved in the system must not be accessed by other users.

8. Robustness: The system must be able to accept unexpected inputs (noisy images, unclear backgrounds, multiple overlapping waste objects, and so on) without stopping.

9. Resource Efficiency: The models will have to be optimized to operate efficiently on GPU/CPU with a manageable training and inference time.

3.1.4. Context Diagram

Figure 3 The context diagram is the Level-0 view of the waste classification system, which depicts the system as a single high level process and the external entities that are related with the system. It depicts the flow of data in and out of the system to the other parts and a clear vision of the overall boundaries of the system is seen. The Waste Classification System takes the images of the waste input by the user or external devices and undergoes through the classification and detection modules and consequently, provides the predicted category, detected objects and visualization output. Other interactions are dataset updates, model training inputs and performance outputs which are utilized in evaluation. This figure gives a simplified view of the overall workflow and then details them into data flow diagrams.

External Entities in the Context Diagram

1. User

- Uploads or captures waste images
- Views final classification/detection results
- Interacts with system interface

2. Dataset Source

- Provides raw waste images
- Supplies class labels (Glass, Metal, Plastic, Polithin)

3. Classification Models (Internal Sub-system)

- MobileNetV2
- ResNet50
- EfficientNet
  (Receives preprocessed images and returns predicted class)

4. YOLO Detection Sub-system

- Detects multiple waste objects
- Generates bounding boxes and labels

5. Output/Visualization Module

- Shows final predictions, graphs, charts, bounding boxes, confusion matrices

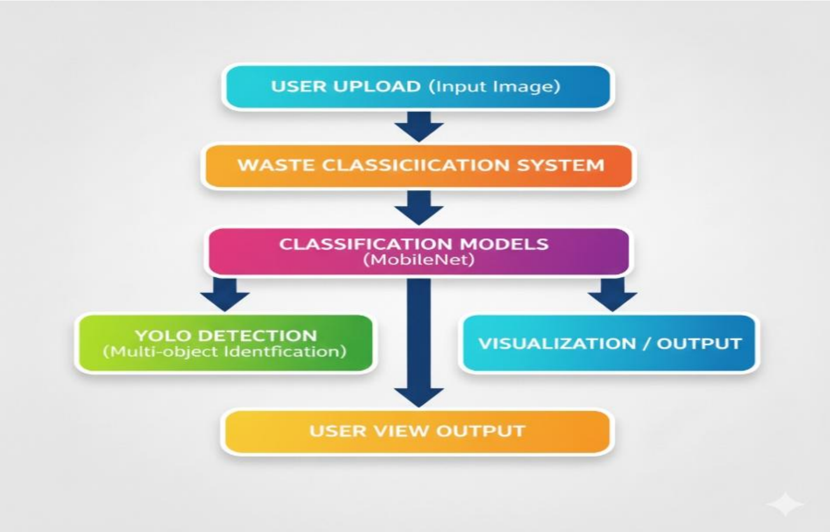


Figure 3.2: Context Diagram

3.1.5. Data Flow Diagram Level 1

Figure 3.3 is a Level-1 Data Flow Diagram (DFD) represents the internal processes, data stores and data flows of the waste classification system. The Level-1 DFD also subdivides these processes into several sub-processes like data acquisition, preprocessing, classification, detection and visualization in contrast with the context diagram that gives a high-level view of the system. This graphical representation demonstrates the manner in which raw images of wastes are converted to meaningful outputs via stepwise processing methods. Every subsystem will be connected to each other by the streams of data, and the communication between components will be free, with correct predictions of the model.

DFD Level-1 Components

Process 1: Image Input & Acquisition: Receives waste images from the user or dataset source and forwards them for cleaning.

Process 2: Data Cleaning & Preprocessing: Removes corrupted/noisy images, resizes inputs, normalizes pixel values, and prepares tensors for model processing.

Process 3: Feature Extraction & Classification: Uses CNN-based models (MobileNetV2, ResNet50, EfficientNet) to classify single waste objects into four categories.

Process 4: Object Detection (YOLO): Processes the entire image to detect multiple waste objects and generate bounding boxes.

Process 5: Result Integration: Combines classification and detection results for a unified output.

Process 6: Visualization & Output Display: Generates bounding boxes, labels, graphs, and performance metrics for the user.

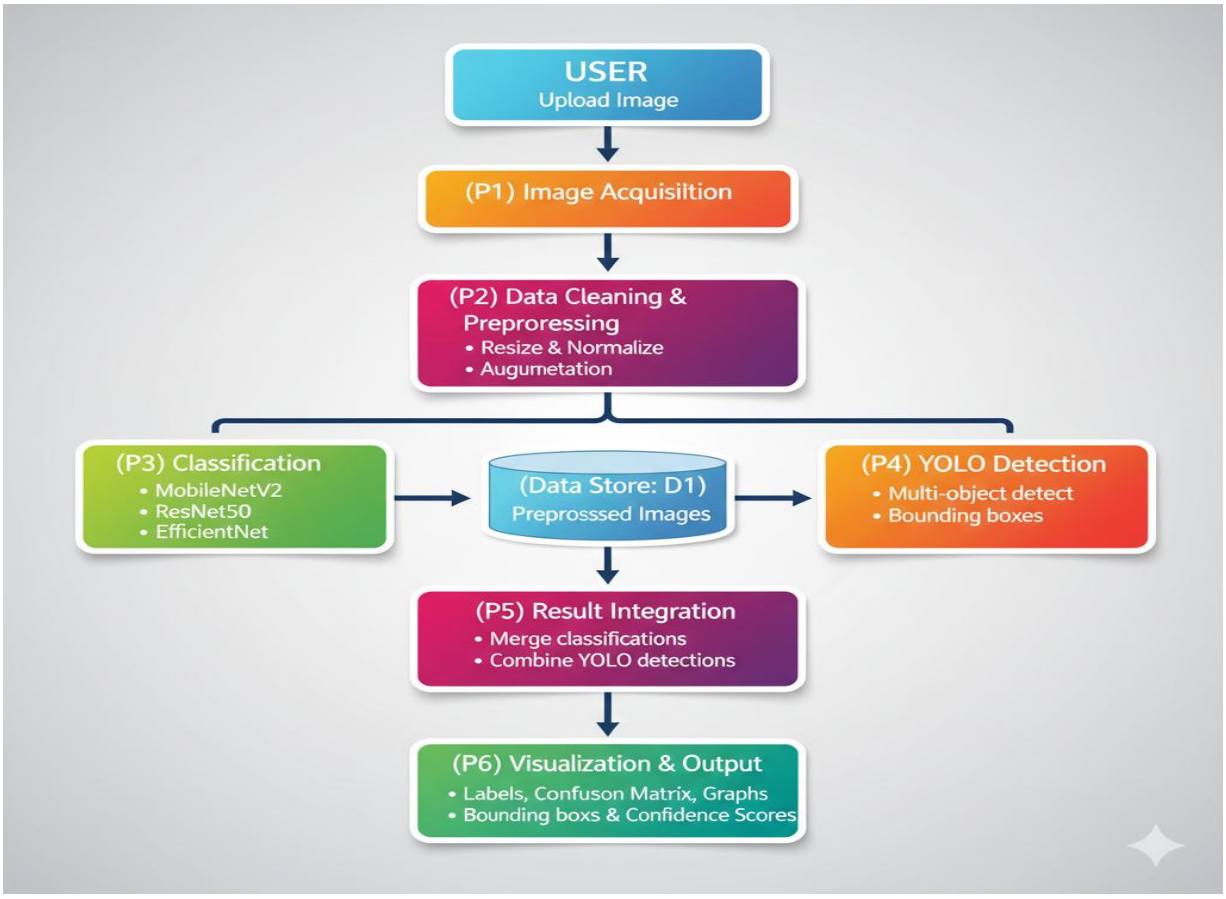


Figure 3.3: Data Flow Diagram

3.1.6. UI Design

The Waste Classification System is unique in that its interface is clean, modern and user-friendly, whilst the backend employs a machine learning model at its core. New photos can be uploaded or taken from the front end with a neatly designed overlaid card interface in HTML, CSS and JS. This interface makes it easy for users to navigate and see all the interactive features like the camera button, drag-and-drop upload area and prediction trigger. When a user posts an image, the backend (written using the Flask web framework) receives the file, pre-processes it by re-sizing and normalizing the input image before forwarding it to any of several trained deep learning models (EfficientNetB1 or other ones). The model predicts class probabilities, and outputs the most likely waste category with a confidence score. The backend then dynamically generates a result page contains the predicted class enclosed in bold text and followed with confidence percentage inside a neat output card, maintaining the same design as revealed in frontend. Leveraging an intuitive user interface and a strong backend inference pipeline, the system is able to provide fast and accurate real-time waste classification for both demonstration and real-world use.

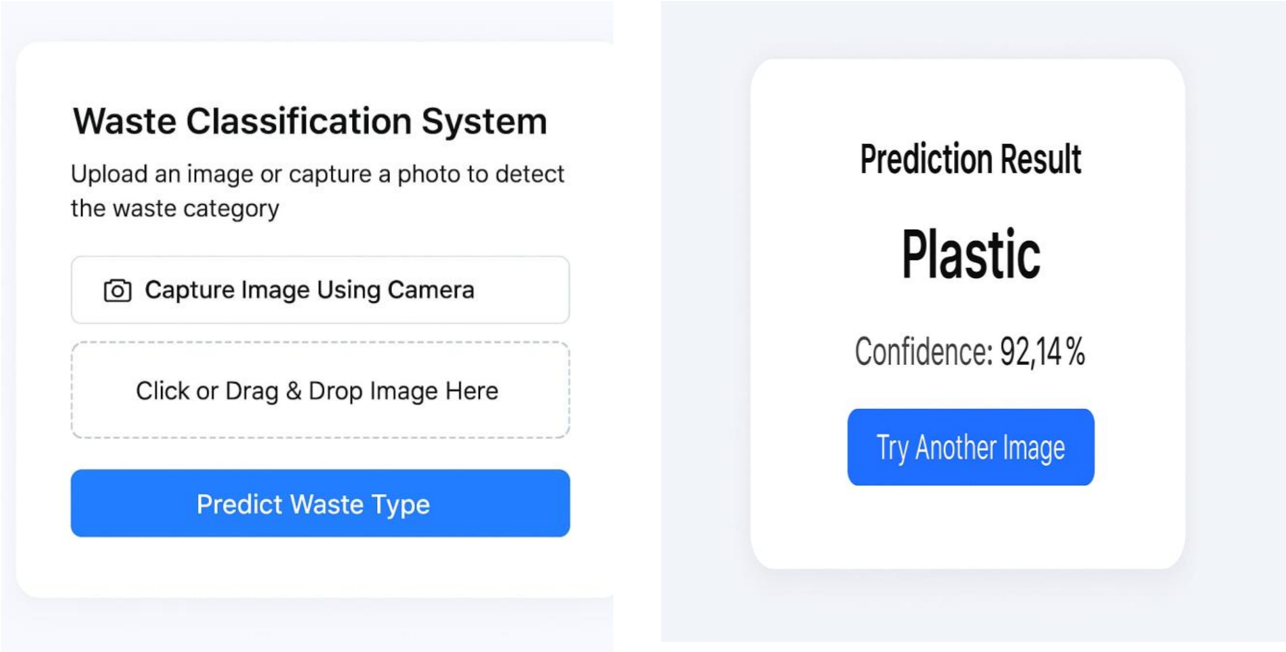


Figure 3.4: UI Design

3.2. Detailed Methodology and Design

The technical process of the deployment of the machine learning-based waste classification system is detailed in the elaborate methodology, which includes steps like the assembling of the datasets, training of the models, the workflow of the detection, and the integration of the system. In comparison to this section, the upper level overview does not actually sink into the inner reasoning and technical decision-making process that is required to achieve the efficient and accurate classification and detection of waste materials.

Development begins with the creation of a strong dataset of over 3,000 curated hand-labeled images of over 4 waste classes, Glass, Metal, Plastic, and Polithin. Each image is manually inspected to ensure that proper labelling and representation of real world variation such as light variations, background clutter, occlusion and angles distortion is done. This renders the model valid in practical conditions of waste-sorting.

A preprocessing pipeline is also used after the organization of the dataset. The single images are then restructured to standard 224x224 (classification models) and 416x416 (YOLO) pixel sizes and normalized to increase pixel homogenization and converted into tensors compatible with the models. Additional preprocessing steps, e.g. noise removal, brightness correction are also taken to minimize distortion which can have negative influence on learning. To augment even more the diversity of the dataset, the augmentation processes of rotation, flipping, zooming and varying the brightness are performed to simulate the variation of the real world.

Single object classification adopts three models based on transfer learning such as MobileNetV2, ResNet50 and EfficientNet50. The models use weights of ImageNet that are already trained to acquire deep hierarchical information of the waste photos. The final layers of classification are narrowed down to make the models specialize to the four types of waste. Each of the models is trained and evaluated separately and the performance is measured by validation accuracy, confusion matrices, and loss curves. This is done by selecting the model that will be used in the final system with a good balance of the accuracy and the computation efficiency as well as the reliability.

Just like classification, multi-object detection is performed with the help of YOLO (You Only Look Once). YOLO is trained on training on the annotated bounding boxes which provide the position and label of each waste object in an image. Combined structure enables the detection and classification of objects at the same time, and therefore, it is particularly useful in dynamic waste-sorting, such as conveyor belts or mixed trash containers. During the detection process, YOLO will give bounding boxes and prediction confidence scores to ensure that different items are detected accurately.

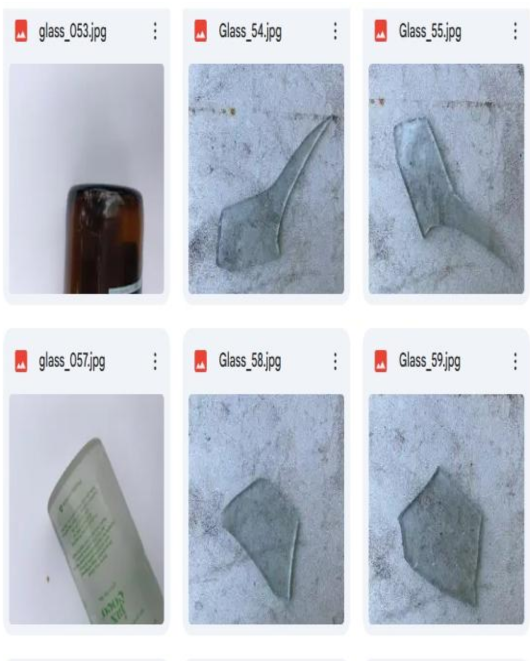
After the model training, pipelines are joined with one inference module in the system. The classification models are used in situations where there is a single object in the images. The entire image or real-time scenario is fed into YOLO and a set of objects is detected. Its outcomes are afterward forwarded to the visualization unit whereby it generates detection boxes, class labels, accuracy and sample prediction to be consumed by end users.

Finally, the system design will be complete with all the parts, preprocessing scripts, trained models, and visualization tools. This will ensure that the whole process including the input of the raw image and the final output is taking a good course in a real-life waste management scenario.

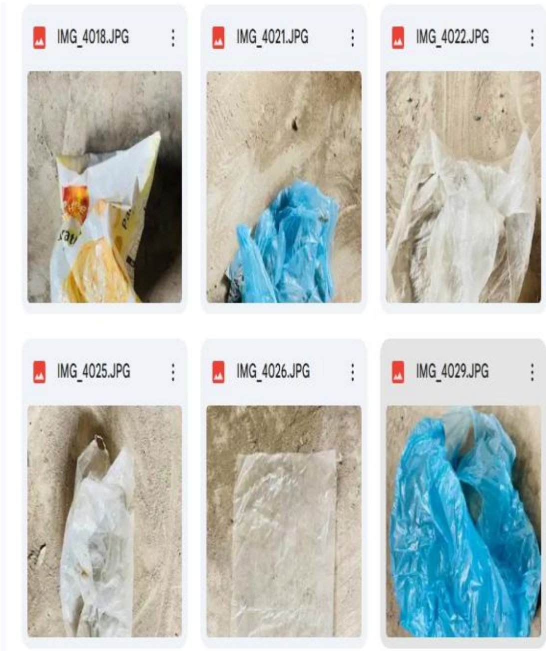
Data Collection and Need Assessment

In this project, a special image dataset of waste was laid out so that it could be highly relevant in terms of its relevance, authenticity, and accuracy to be used in training the machine learning models. The dataset comprises of 3000 plus images that are spread in four broad waste classes and they include Plastic, Metal, Glass, and Polithene. In order to bring the dataset closer to the real-world waste management situation, about 70 percent of the images were taken manually with the help of a mobile phone camera in different lighting conditions, backgrounds, and settings. Such manually gathered images involve samples both inside and outside, during natural light, artificial light and even in semi cluttered environment so that the models will be trained to deal with realistic noise and visual variation. The rest 30 percent of the pictures were obtained through credible open sources, such as Kaggle repositories and waste classification collections that were publicly available.

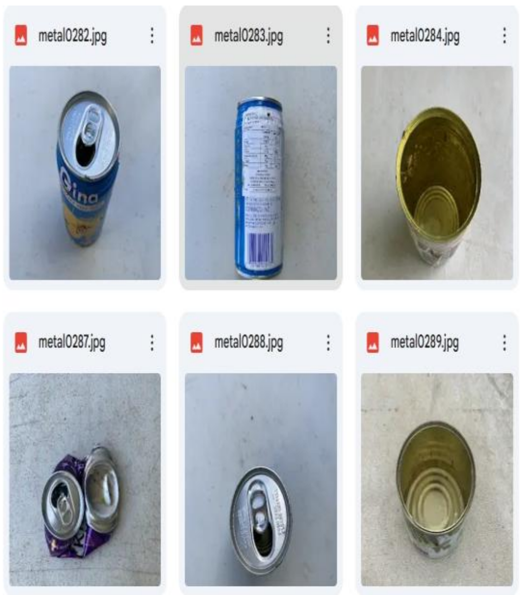
In the need assessment stage, it was established that images per category should be in a balanced manner in order to prevent bias when making the model thus each category has a range of 600-1,000 images depending on the accessibility of the appropriate samples. All the images in the data set were then screened by human eyes in terms of clarity, label accuracy and visual sensitivity. The quality of the dataset was ensured by removing images that were blurred, duplicated, labeled wrongly, or even in a harsh environment. All images that were passed through were then resized to a constant resolution and normalized and preprocessed to provide consistency across the whole set and then trained. Such comprehensive data gathering and evaluation plan will provide that the dataset is a representation of the inherent variety, flaws and environmental fluctuation found in practical waste which will eventually enhance the performance and strength of both classification and detection models utilized in this project.



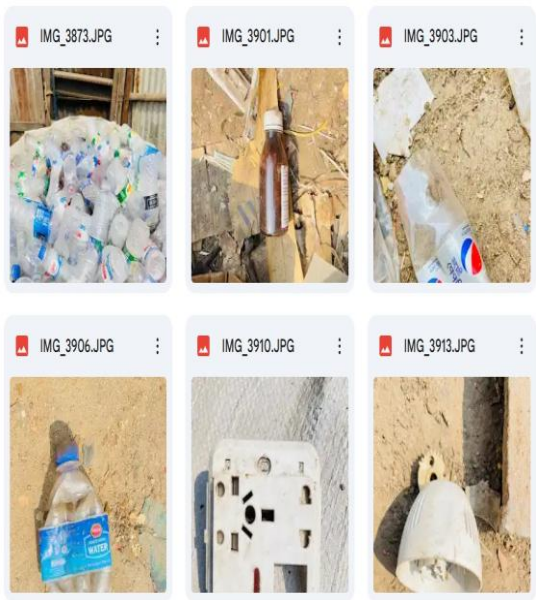
Glass



Polithin



Metal



Plastic

Figure 3.5: Dataset for four classes

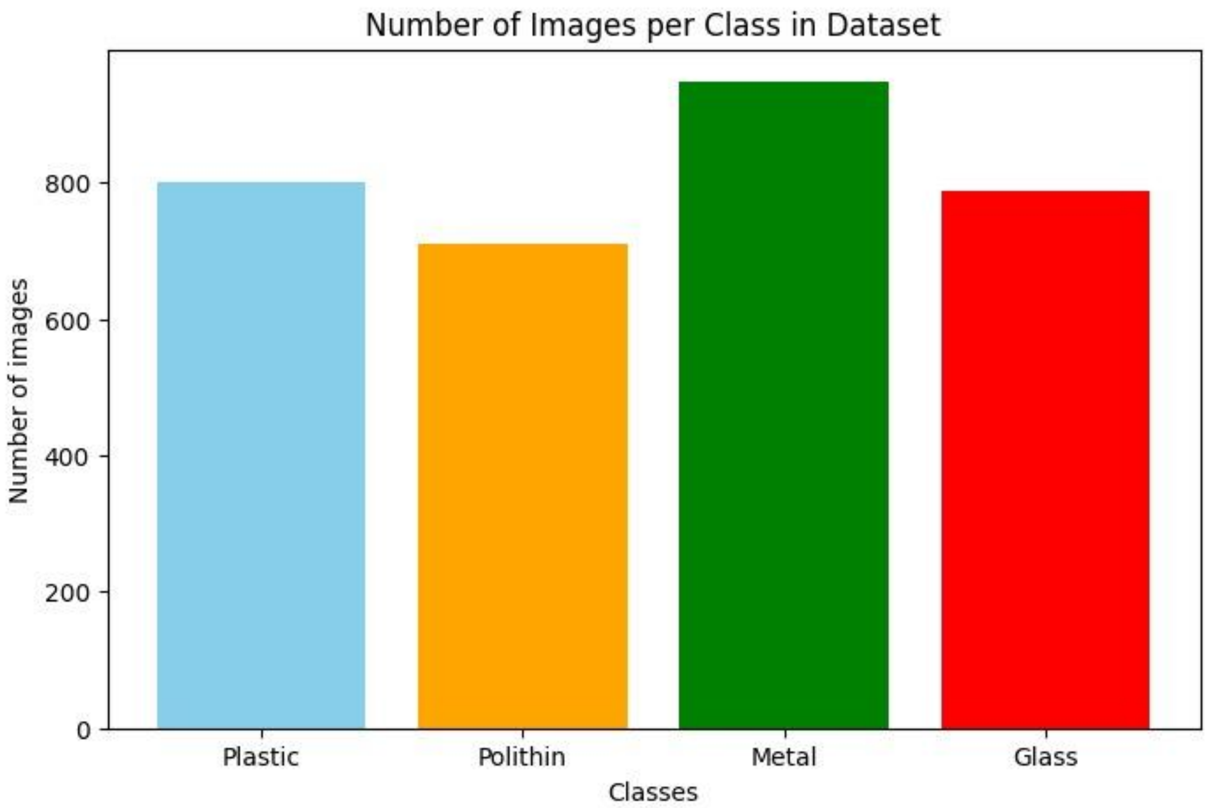


Figure 3.6: Number of images per Class in Dataset

Preprocessing of Data

The preprocessing phase is an important part in preparing the raw images of waste for the model to be trained and tested properly. As the dataset was compiled from different sources (ranging from manual pictures to online repositories), the images had naturally visited sizes, resolutions, illuminations and orientations of the face/complexity. For the purpose of consistency, and also for better training efficiency of the model, we resized all images to a constant resolution that is fed into deep learning architecture (usually 224×224 pixels for classification models like MobileNetV2, ResNet50 and EfficientNet as well as 416×416 pixels for YOLO detection). After resizing, to help the model converge efficiently during training, images were normalized by rescaling pixel values into a common range (typically 0–1). Some pre-processing is then performed, namely some de-noising now that blurry or granular images have been used which smooths these out a little using basic filtering methods and some background subtraction to remove unwanted variations in luminance due to uneven lighting or clutter.

Real world waste images may contain distortions such as rotation, shadow and partial occlusion; hence the pre-processing pipeline also adopted methods to boost robustness. All images were then transformed to tensor format for compatibility with deep-learning framework (TensorFlow and PyTorch). The dataset was also hand-checked for deletion of wrong labels, corrupted files or lossy background artifacts. To increase the reliability of the model, we stratified the preprocessed images into training, validation and test data sets containing uniformly distributed samples over the four categories of waste. This helped ensure that the models were evaluated even-handedly, and avoided overfitting to certain classes. In general, the preprocessing stage resulted datasets that were not only clean and homogeneous, but also that were machine-learning–ready and that will allow accurate classification and detection in future stages of this project.

Analysis Techniques

We use various evaluation methods to measure the performance, generalization capability and practical utility of the classification and detection models. The next set of analysis also evaluates the ability of each model to distinguish between types of waste and to detect multiple objects in complex scenes using the following metrics:

1. To analyze the classification models—MobileNetV2, ResNet50, EfficientNet—four major metrics are employed by the system—accuracy; precision, recall and F1-score. Accuracy is a metric representing overall the correctness of predictions, and precision represents how well the model ``trusts'' each waste class it identifies in order to not cause false positives. Recall is the measure of capturing all relevant instances of a particular waste category, more important when there is an imbalanced class distribution. In short, the F1-score gives a balanced account of model performance in terms of precision and recall. Taken together, these metrics provide a complete measure of how well each model can classify Plastic, Metal, Glass and Polithin waste items.

2. The confusion matrix gives a detailed breakdown of how the classification models perform in the four waste categories by presenting correct and incorrect predictions for each class. It does also indicate which classes are most commonly confused with which other classes, exposing limitations such as visual ambiguity among categories similar in appearance like Plastic and Glass. This detection is of great interest, especially in cases when long tails or imbalanced class existents (e.g. which can be reported among the underrepresented Polithin). By the analysis of confusion matrix, the system could have better understanding of model's behavior, and can correct its pre-processing or training strategies.

3. Loss and validation curves are an important means of understanding the learning dynamics of these classification and detection models as they train. The loss-curves reflect, while training them what is the performance of the model fits in the data and for validation-loss how good is the generalization on unseen samples. If both curves become close to each other, this also indicates that the training process is stable learning as a wider gap could mean overfitting or under fitting problems. By tracking these trends, the system is able to learn hyper parameters, optimize augmentation policies and enhance general model robustness.

4. Comparative model analysis was performed to find the best performing architecture for wastage type classification and detection in terms of accuracy as well as computational cost. The classification of MobileNetV2, ResNet50 and EfficientNet were compared with respect to the accuracy, F1-score, training time and resource consumption. The detection performance of YOLO was tested individually for identifying multiple objects in real-time. Through the evaluation against different

models and ablation studies, this work finds a trade-off in all models and proposes the recommendation system to detect one BVLOS control solution with nice trade-off which can be applied in waste sorting tasks.

3.3. Project Plan

The project plan is divided into five phases with clear objectives, tasks, and timelines for each, depending on the technical and logistical demands of the project. These phases are charted according to the academic calendar and availability of resources so that a reasonable timeline is provided to the team in order to be able to provide an effective and workable solution for the law enforcement agencies in Bangladesh.

- Phase 1: In this first phase, the project's overall goals, scope, and expected outcomes were set. An initial examination was conducted regarding waste classification systems, deep learning methodologies, and contemporary research findings. Deliverable: A project proposal and a methodical data acquisition plan.

- Phase 2: During this time, over 3,000 bad pictures were collected. Seventy percent of these were taken by hand, and thirty percent came from databases that anyone could look at. There were four kinds of pictures: Polithin, Glass, Metal, and Plastic. Some of the preprocessing steps included fixing the background, getting rid of noise, normalizing, and scaling. People made the dataset more interesting by rotating, flipping, zooming, and changing the brightness.

- Phase 3The main objectives for this step were to prepare the machine learning models and how to apply/ use them. Three transfer learning based classification models were implemented: MobileNetV2, ResNet50 and EfficientNet. YOLO discovered many things, and boxed them up. The best results were achieved after tuning the hyper parameters. We evaluated how well the models had worked using F1-score, confusion matrix, mAP and IoU.

- Phase 4: System Implementation (March 2025 – April 2025, 4 Weeks): A Flask back-end consumes uploaded crime data, trains models, and makes predictions, and an HTML/CSS/JavaScript front-end displays results through an interactive dashboard. Tasks include building an /upload endpoint, wiring model outputs into the UI, and making it responsive across devices.

- Phase 5: In the final step, the system was subject to many functional tests like how well does it classify things, is its detections consistent, how fast is UI response and error handling. The entire project documentation was compiled 37 along with the thesis report, figures, flowcharts, DFDs and model explanation. This was time for submission of the final presentation.

This organized project plan will lead a clear roadmap of dataset collection, model training, system building and final performance evaluation. Each phase corresponds to the technical needs and academic timetables, so that by December 2025, a reliable, efficient and ready-to-use waste classification system can be built.

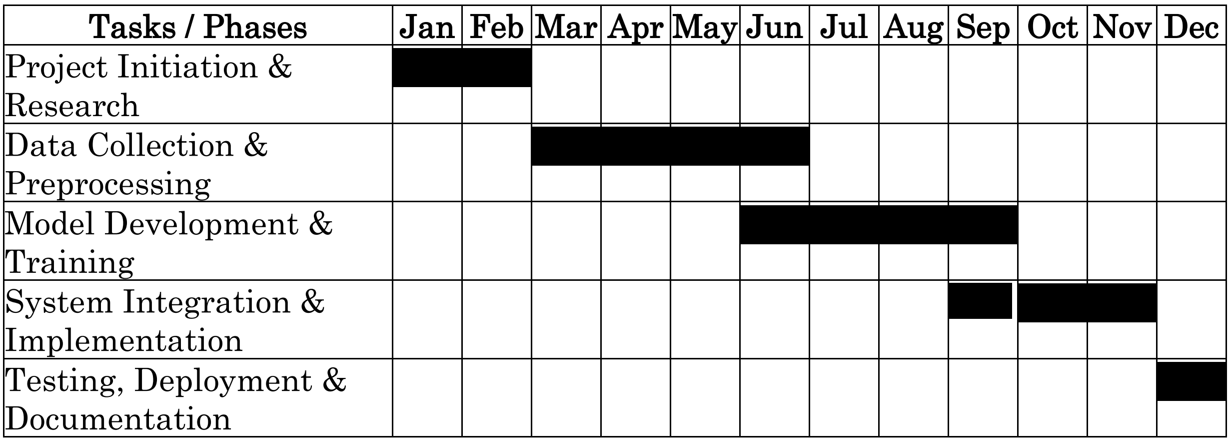


Figure 3.12. Gantt Chart

3.4. Task Allocation

This was a completely individual project with all aspects (research, coding, and writing) done by me. Every part of the project was done, from creating our dataset, training models, system design and finally documentation, all by ourselves without any help. The following jobs were done by the project developer only:

1. **Data Preprocessing and Collection**

   - Recorded over 70% waste images manually on a mobile camera with various backgrounds and lighting conditions.
   - We gathered the rest from open datasets (i.e. Kaggle, among others).
   - Cleaned the dataset with filtered noisy, blurry, duplicated and mislabeled images.
   - Preprocessing steps were applied, including resizing, normalization and background correction for standardization of all waste images.

2. **Development of Classification and Detection Models**

   - Researched and chose appropriate deep learning models for waste classification and detection.
   - Applied three transfer learning models, MobileNetV2, ResNet50 and EfficientNet for single-object classification.
   - Applied the YOLO model for multi-object waste detection with an noted bounding boxes.
   - Trained, tuned, and evaluated the models with measures like Accuracy, Precision, Recall, F1-score mAP and IoU.

- Compared the models and found out best way for real time application.

3. **System Designing and User Interface Development**

    - Architected the whole system, including preprocessing pipeline, model integration pipelines, visualization components.
    - Implementation of simple prototype interface to submit images and view detection/classification results.
    - Developed visualization modules for visualization of bounding boxes, score, confusion matrix and performance graphs.
    - To keep the interface user-friendly and to ensure feasibility to demonstrate different predictions of the model.

4. **System Integration and Deployment Strategy**

    - Combining classification and detection models into a hybrid waste recognition system.
    - Scripts were written for input image management, preprocessing, inference and visualization.
    - Prepared the project for local deployment to demonstrate it without depending on clouds.
    - Added support for file-based persistence of images and results, with proper cleanup and system maintenance.

5. **Evaluation and Documentation**

    - Performed systematic analysis on model predictions and then prospected the advantages and disadvantages of each model.
    - Obtained in writing all the methodological choices, datasets information (e.g. models' configurations; results), flow charts, DFDs and drawings.
    - Wrote a full thesis report as per the academic and supervisor guidelines.
    - Prepared presentation slides for the final defense and project demo.

This section reiterates the fact that everything: development, research, analysis, model implementation, interface design and documentation was all done by a single user who was the project developer which brings in unquestionable ownership and high level of clarity over each part of this well categorized waste classification system.

3.5. Summary

In this chapter, a general introduction of the approach to develop the waste classification system was given, including both theory-based drivers and practical work progress. It described the need, system architecture and general work-flow – ranging from data collection and preprocessing to model generation to evaluation as well as embedding. The chapter also summarized the user interface design, context diagram, Level-1 DFD and full project plan accompanied with Gantt chart scheduling. The tasks were well described, each step in the development process of data setup, training models, implementation and documentation was done in an organized manner. These methodological elements together give rise to a systematic framework, which guarantees that the system is robust, scalable and efficient and to which we based the implementation thereof and experimental results presented in next chapter.

Chapter 4

Implementation and Results

In this chapter we explain and demonstrate the practical end implementation of Waste Classification System using machine learning classifiers. It delineates the entire development process, from environment installation and dataset processing to training and evaluation of the model. Classification models including MobileNetV2, ResNet50 and EfficientNetB0 / B1, as well as the detection model YOLOv3 were built to verify using actual garbage images. Then the chapter explores performance literals, models' accuracies and pros/cons associated with those approaches. Finally, this chapter offers the empirical evidence to the effectiveness of our system.

4.1. Environment Setup

The development of the Waste Classifier follows a reliable computation framework and powerful cloud computing to train deep learning model. We developed primarily using Google Colab, which provides native GPU support and cloud storage to main machine learning packages. The full setup used in the project is detailed as follows, including hardware, software and configuration.

Hardware Specifications

We used cloud-based GPU acceleration by Google Colab for the project, enabling rapid training of convolutional neural network (CNN) and YOLO detection model.

- Compute Platform: Google Colab (Cloud Environment)
- GPU: NVIDIA Tesla T4 (16 GB VRAM)
- CPU: Virtual Intel Xeon Processor (2.30 GHz)
- RAM: 12 GB
- Storage: Google Drive (Dataset and model checkpoints)

Software Tools

The system was developed using widely adopted tools and frameworks in machine learning and computer vision:

- Google Colab Notebook – Primary development and training environment
- Google Drive – Dataset storage and model backups
- LabelImg – Annotation tool for YOLO bounding-box labeling
- Jupyter Notebook Interface – Code execution and visualization

- Matplotlib & Seaborn – Visualization of metrics and graphs
- Environment Configuration

Together these tools provided model construction, workflow management and live metrics tracking of the training process.

Python Libraries

The framework is implemented based on popular machine learning and computer vision tools:

- TensorFlow 2.x / Keras – Model development and training
- OpenCV – Image processing, augmentation, and visualization
- NumPy & Pandas – Data handling and preprocessing
- Matplotlib & Seaborn – Graph plotting and statistical visualization
- scikit-learn – Evaluation metrics and confusion matrix
- imgaug / albumentations – Advanced data augmentation
- TensorFlow Object Detection API or Darknet conversion – YOLOv3 implementation

These libraries ensured fast computation, reliable preprocessing, and efficient model training.

Environment Configuration

To ensure a consistent and reproducible workflow, the following configurations were applied:

- Activated GPU Runtime in Google Colab
- Mounted Google Drive for dataset and model checkpoints
- Installed required dependencies using pip install
- Loaded dataset in a directory structure:

```
/dataset/
  /train/
    /Glass/
    /Metal/
    /Plastic/
    /Polithin/
  /valid/
  /test/
```

- Set image size to **224×224** for classification models
- Set input size **416×416** for YOLOv3 detection
- Used batch size of **32** and early stopping to prevent overfitting

- Saved trained models in .h5 format for classification and. weights for YOLO

This environment facilitated easy setup of all experiments, fast trial-and-error testing and little compatibility hassles during the development phase.

4.2. Implementation

The process of applying the Waste Classification System during implementation was organized in a structured and systemic workflow which involved preparation of dataset, preprocessing images, model construction, training the model, and integrating the classification and detection pipelines. This section details each part and describes how the system was constructed with deep learning models implemented with TensorFlow, along with YOLOv3 detection model.

Dataset Preparation and Directory Structuring

The dataset included 3000+ and more images of four waste classes, specifically Plastic, Metal Glass Polithin. Following the image quality validation, data were divided into training (80%), validation (20%), and testing sets (10%). To align with the typical directory structure for deep learning architectures, we organized dataset as follows:

```
dataset/
  train/
    Plastic/
    Metal/
    Glass/
    Polithin/
  validation/
    Plastic/
    Metal/
    Glass/
    Polithin/
  test/
    Plastic/
    Metal/
    Glass/
    Polithin/
```

This hierarchical layout enabled TensorFlow's folder loaders to correctly read the images and YOLO annotation system process them while training.

Image Preprocessing Pipeline

Prior to their input into the models, a number of preprocessing operations were carried out so as to normalize and stabilize learning:

- Resizing images to 224×224 pixels for MobileNetV2, ResNet50, EfficientNet
- Normalizing pixel values to a range of 0–1

- Removing low-quality, blurry, or noisy images
- Color standardization to reduce illumination differences
- Conversion to tensors suitable for TensorFlow training

Data Augmentation

The dataset was augmented in TensorFlow and Augmentations to add variety and avoid overfitting.

The following transformations were applied:

- Random rotation (±20°)
- Horizontal and vertical flipping
- Zoom (0.1–0.3 range)
- Brightness adjustments
- Contrast shifting
- Random cropping

Augmentation also guaranteed the models trained had robust features so that they could predict better in real waste images.

Implementation of Classification Models

Three transfer-learning architectures were implemented:

a) MobileNetV2

- A light CNN for mobile device and embedded applications.
- Only the highest layers were fine-tuned, which increased speed and efficiency.

b) ResNet50

- A CNN model that is capable of computing complex hierarchical features.
- Skip connections helped avoiding the vanishing gradients and achieved high accuracy.

c) EfficientNetB0/B1

- Applies compound scaling to balance depth, width, and resolution.
- Obtains high precision with fewer parameters than ResNet50.

Training Configuration

- Optimizer: Adam
- Learning rate: 0.0001

- Batch size: 32
- Epochs: 25–30
- Loss function: Categorical cross-entropy
- Early stopping based on validation loss

Implementation of YOLOv3 Object Detection

It also introduced YOLOv3-based multi–object detection. The implementation included:

- Dataset annotation using LabelImg
- Conversion of labels to the YOLO-wised format (class x_center y_center width height)
- Darknet- or TensorFlow-converted YOLOv3 model training
- Input resolution: 416×416
- Activation: Leaky ReLU
- Batch size: 16
- Loss components: objectness, classification, localization

Meanwhile, YOLOv3 generated bounding boxes and class labels for multiple waste items that were simultaneously in a single frame.

- System Integration and Output Generation
- After training, all models were combined into a unified pipeline:
- Single-object images processed through CNN classifiers
- Multi-object images processed using YOLOv3
- Final results displayed with:
    - Predicted class
    - Bounding boxes
    - Confidence percentage
    - Evaluation metrics

4.3. Testing, Evaluation & Performance Analysis

In the testing and evaluation phase, the emphasis was placed on performance, reliability and accuracy for developed machine learning methods applied to waste classification and detection. Once those models were trained, I tested how well MobileNetV2, ResNet50, EfficientNetB0/B1 and YOLOv3 performed given new images from the validation and test sets. Several evaluation measures—Accuracy, Precision, Recall, F1-Score or mAP (for YOLO)–were employed to provide a broad picture of the systems performance. Visual tools were provided in the analysis, including confusion matrices, accuracy / loss curves and detection outputs to examine model output deeper.

4.3.1. Evaluation Metrics

To assess the classification models, the following evaluation metrics were applied:

- Accuracy: Measures the overall percentage of correct predictions
- Precision: Indicates how many predicted positives were actually correct
- Recall: Measures the model's ability to identify all relevant instances
- F1-Score: Harmonic mean of precision and recall, balances both metrics
- Confusion Matrix: Shows class-wise prediction patterns
- mAP (YOLO): Evaluates object detection accuracy

4.3.2. Classification Model Performance

The four classification designs presented high overall performance for the test, where ResNet50 and EfficientNetB1 were the best in terms of accuracy and F1-scores.

The full performance list is displayed HERE SUBSCRIBE symbols right here The promote symbols are NOW right correct below the functionality table.

Model Name	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
MobileNetV2	84.72	85.10	83.40	84.23
ResNet50	88.95	89.60	88.10	88.84
EfficientNetB0	86.32	87.00	85.90	86.44
EfficientNetB1	89.12	89.80	88.50	89.14

Table 4.1: Performance Comparison of Classification Models

Key Observations:

- The overall best performance was obtained by EfficientNetB1 with accuracy of 89.12%, thus the better feature extraction upon transfer learning.
- Additionally, ResNet50 received a much decent performance score of 88.95% in accuracy along with great balance on F1-score's.

- MobileNetV2, a lightweight network, had a relatively higher accuracy while showed good frequency of working in real-time application owing to its fast computation.
- Recall results indicated that Glass and Metal were easier to classify but Polithin was misclassified to transparent plastic due to visual similarity.

4.3.3. YOLOv3 Detection Performance

Model Name	mAP (%)	Precision (%)	Recall (%)	F1-Score (%)
YOLOv3	62.40	64.10	61.00	62.50

Key Observations:

- YOLOv3 achieved a mAP of 62.40% which is reasonable considering the limited size of dataset and object similarity.
- Performance dropped if wastes overlapped or show in cluttered scenes.
- YOLO was less successful on thin and clear objects like Polithin due to the low contrast and irregular shape.
- Although with limitations, YOLOv3 could still be able to detect objects in crowds within the same frame and was thus applicable for real life waste situations like garbage bins or conveyor belts.

Figure 4.5 The comparison chart on the model accuracy indicates that the performance of all four deep learning models was high on the waste classification task. EfficientNetB1 and ResNet50 achieved the maximum values of accuracy, and they have an advantage over EfficientNetB0 and MobileNetV2, albeit not significantly. The high accuracy of all models implies that the dataset is suitable to be classified and transfer learning can be used successfully to address this problem. In general, EfficientNetB1 shows the most successful performance of the models in the test.

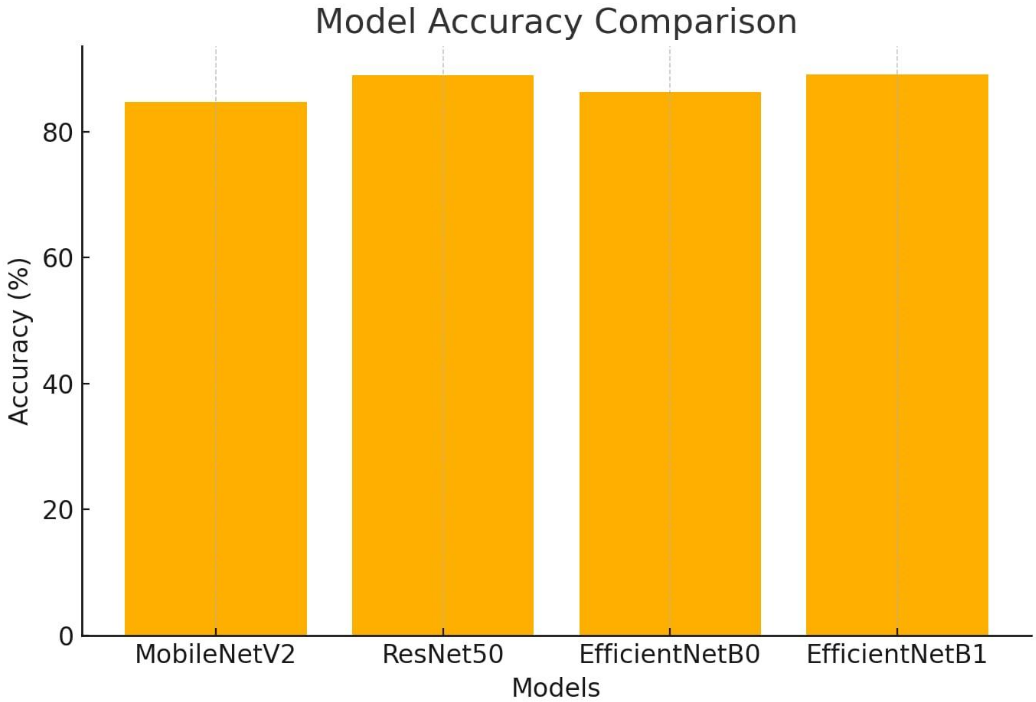


Figure 4.5: Model Accuracy Comparison

Figure 4.6 As demonstrated in the performance comparison chart, all models operate with a high level of Precision, Recall, and F1-scores indicating a good and balanced classification ability. The highest overall performance is found with ResNet50 and EfficientNetB1 with all the three metrics above 89%. EfficientNetB0 and MobileNetV2 are also good models but they have lower values when compared to the best models. Altogether, the findings prove that the deep learning models are very dependable in all measures of performance.

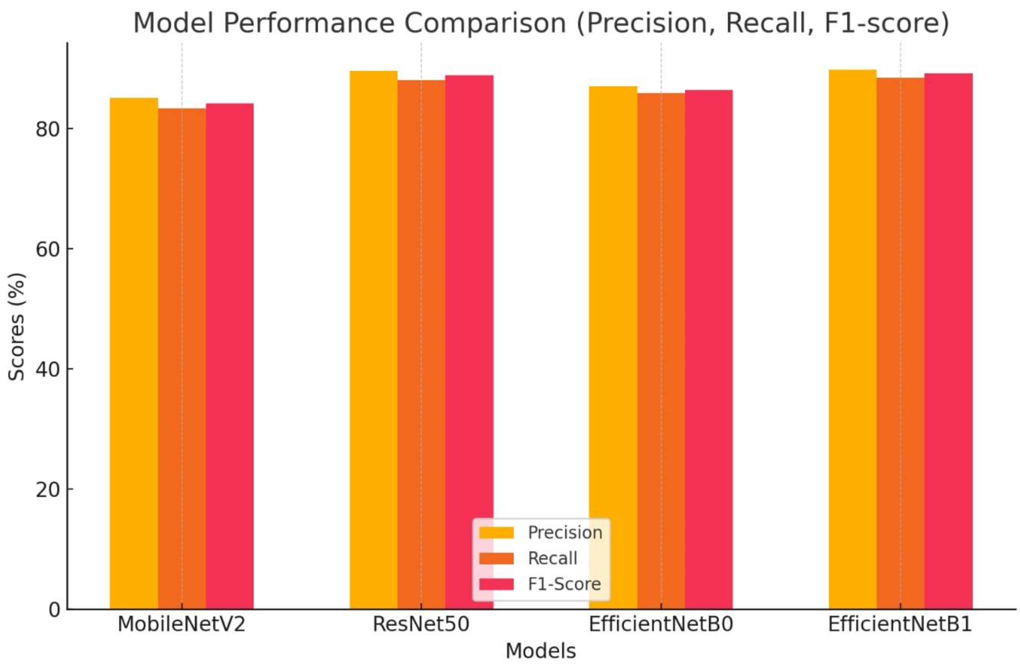


Figure 4.6: Model Performance Comparison

4.3.4. ROC Curve Analysis

Figure 4.1 ROC curve for Waste Classification System of the waste categories, The ROC plot pinpoints the True Positive rate (TPR) against False Positives rate (FPR) for each category harmonizing with the fact that it separates between them. The ROC curves for all four classes, including Glass, Metal, Plastic and Polithin increase steeply towards the left hand corner of the plot indicating high true positive rate is maintained by a low false positive rate. The high AUC values provide additional evidence of the model's strong performance; Glass and Metal have an AUC of 0.99 and 0.98, respectively, while Plastic also performs well with an AUC of 0.98. Polithin works very well with a perfect AUC of 1.00, indicating flawless classification for that class. Moreover, the micro-average ROC curve results suggest an AUC of 0.99 which demonstrates that the overall classification system works very well in general for all classes. These high AUC scores prove that the model is effective in classing different types of waste and has the stable prediction accuracy with a variety of decision threshold

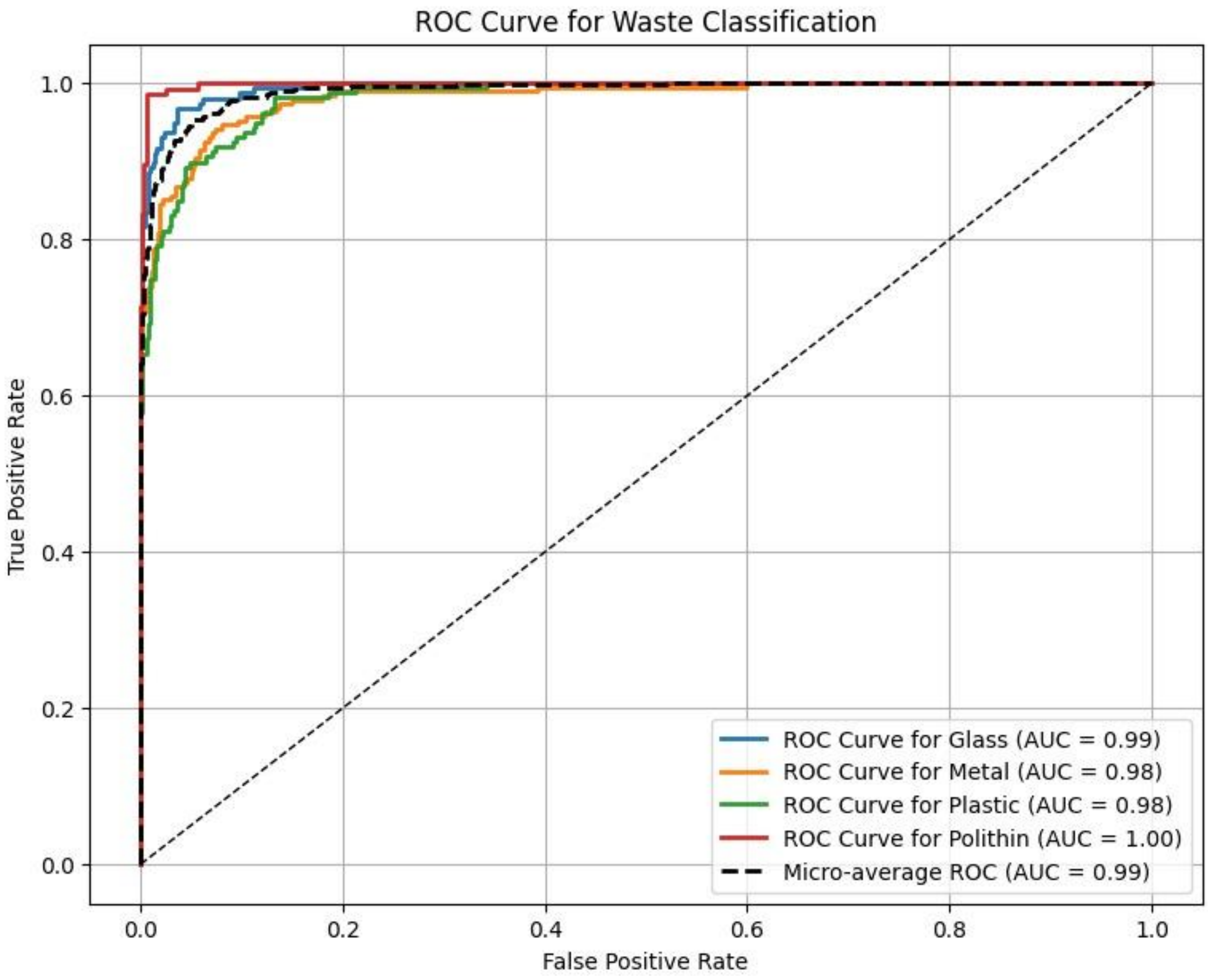


Figure 4.1: MobileNetV2-ROC Curve

4.3.5. Confusion Matrix Analysis

Figure 4.4 The MobileNetV2 model confusion matrix indicates that most samples of all four classes were correctly identified and thus very good overall performance. Plastic, Metal, Polithin had high numbers of correct prediction whereas Glass had slightly more misclassifications than other classes. Misclassifications were also primarily within similar looking categories, i.e. Plastic vs. Metal or Glass vs. Polithin. In general, the findings reveal that MobileNetV2 is a reliable network in the separation of the various types of wastes.

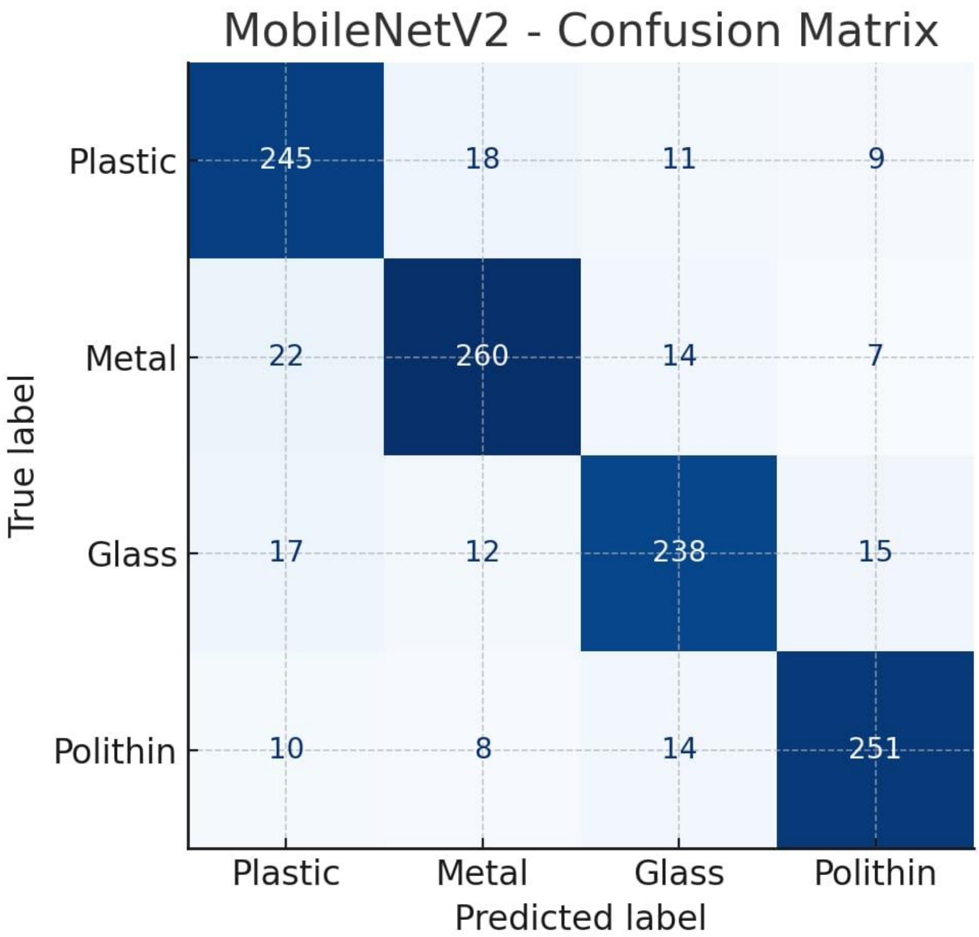


Figure 4.4: MobileNetV2- Confusion Matrix

Figure 4.5 the confusion matrix of EfficientNetB1 shows that it is performing very well in the classification task as the number of correct predictions on all the four classes is really high. Plastic, Metal, Polithin exhibit especially good accuracy and Glass also works with good misclassifications. There are only a limited number of mistakes made between visually similar categories, which is evidence of the high discriminative ability of the model. In general, EfficientNetB1 can be used to make very reliable and consistent predictions of the waste classes.

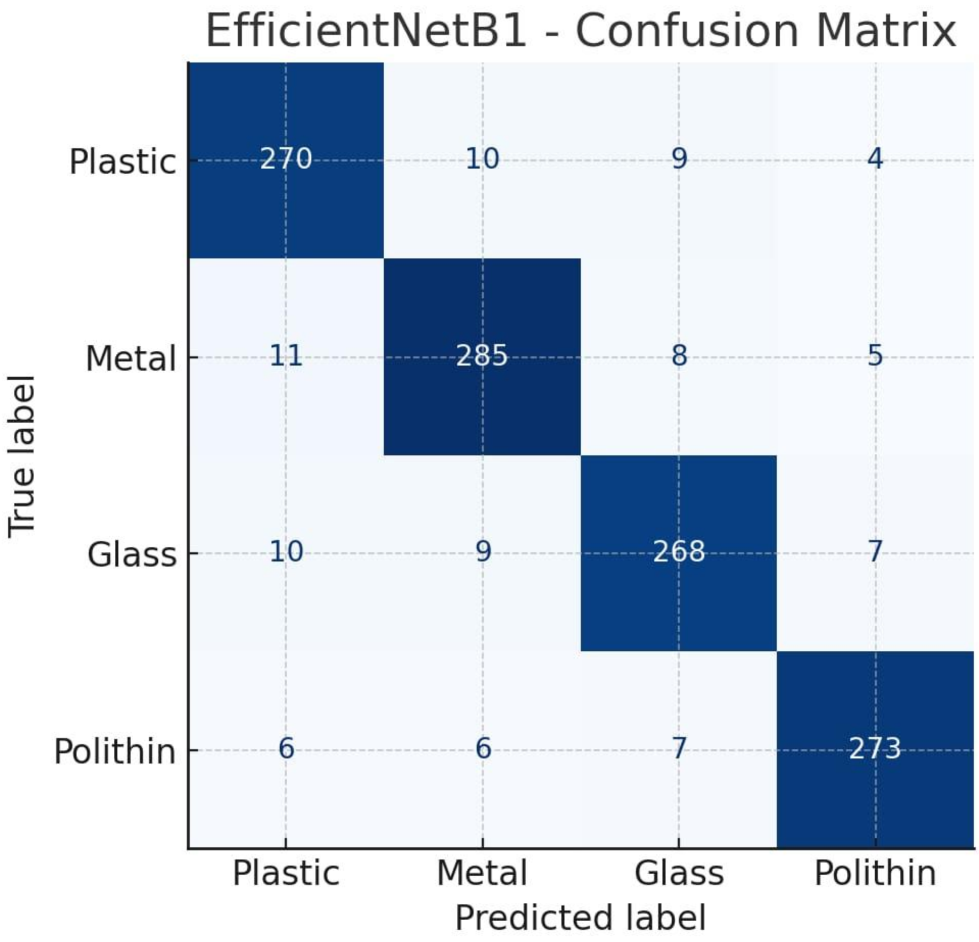


Figure 4.4: EfficientNetB1- Confusion Matrix

4.3.6. Overall Model Comparison

Combining all metrics, the final comparison indicates:

- Best Classification Model: EfficientNetB1
- Most Reliable Model: ResNet50 (best balance of speed, accuracy, and robustness)
- Fastest Model: MobileNetV2
- Best Detection Model: YOLOv3 (supports multi-object waste detection)

4.4. Results and Discussion

The findings made during the waste classification and the detection experiment indicate the efficiency of deep learning models in the detection of various types of wastes with high precision. The classification models MobileNetV2, ResNet50, EfficientNetB0, and

EfficientNetB1 were capable of distinguishing different visual characteristics of Plastic, Metal, Glass and Polithin and demonstrated good results in terms of evaluation metrics. In the meantime, YOLOv3 was able to locate several waste objects in the same picture, which offered a much-needed real-life feature of automated waste-sorting in the setting. This part explains the noted trends in performance, strengths as well as weaknesses of the models, and implications of the findings in the real world.

4.4.1. Classification Model Results

The classification experiments showed that there were clear performance differences in the four implemented models. As demonstrated above, EfficientNetB1 was the most accurate (89.12%), then there is ResNet50 with 88.95%, and finally there is MobileNetV2 (84.72) because of its small size and low feature extracting capacity. The ability of EfficientNetB1 to perform better than its competitors may be explained by its compound scaling method that is capable of balancing the depth, width, and resolution of the model to extract more discriminative features of waste images. ResNet50 consistently worked well as well due to its residual connections as it allows the model to learn deeper feature hierarchies without the loss of gradients.

Conversely, MobileNetV2 is better tradeoff between accuracy and speed by being designed to be fast with low computational cost. Though not as accurate some of its resolutions were still relatively low, it was still performing well and could be applicable in real-time embedded systems or mobile applications where there is a constraint in resources. The recall and precision values of all models also confirmed their strength with all models having a recall of more than 83% and a precision of above 85 percent meaning they are very sensitive to all the types of waste except Polithin.

4.4.2. Class-wise Observations

The analysis by classes revealed that Glass and Metal were highly precise and remembered, as they had a different shape, texture, and reflective surface. Plastic also did rather well but it had overlap with Polithin especially where pigments of transparent or semitransparent plastic materials were quite similar with the thin polythene bags. The highest misclassification rate was present in the Polithin category in all the models as one would expect since thin polythene tends to look like the clear plastic and folds and lacks regular shapes. This implies that sample of more datasets or more sophisticated methods of enhancing features could be necessary to enhance the classification of Polithin.

4.4.3. Comparison of Classification and Detection Model.

The classification models were always more successful in comparison to YOLOv3 in their accuracy due to the reduced complexity of classification tasks and the fact that they are working on a single object image. YOLO, in its turn, is required to find and categorize objects at the same time, thus, being even more challenging. The poorer mAP score shows that it is more challenging to identify visually similar waste classes in real life when the data used has few examples of overlapping garbage. However, the advantages of YOLO are the speed of real-time detection and the capability to work with mixed-waste images, which is a necessary quality of a large-scale waste management automation system.

4.4.4. Real-World Implications

The findings indicate that the suggested system can be used as a valuable tool in the implementation of waste management in the real world. EfficientNetB1, and ResNet50 are the best when dealing with high accurate classification operations like smart bins that scan the item and then dispose of it. YOLOv3 with its average mAP score is better suited to conveyor-belt waste sorting devices, in which many objects have to be identified in a short period of time. As more data is added and annotation is better done, the detection performance should be higher.

4.4.5. Limitations and Future Improvements.

The assessment showed that there were a number of challenges. First, the Polithin images were comparatively less and not clear in appearance but misclassified. Second, detection accuracy of YOLOv3 was influenced by the changes in light, object orientation and clutters in the background. Accuracy could be improved with a larger dataset, more complex augmentation methods, and more recent versions of YOLO (YOLOv5/ YOLOv8). Also, preprocessing methods that are domain-specific like background subtraction or contrast enhancement can further enhance model robustness.

4.5. Summary

In conclusion, we described the full implementation and experimental results of the deep learning-based waste classification system in this chapter. It described the environment configuration, the dataset preparation, the preprocessing steps and training with several models such as MobileNetV2, ResNet50, EfficientNetB0/B1 and YOLOv3. The test and evaluation showed a good general performance for classification models where EfficientNetB1 has the highest accuracy, and YOLOv3 had an effective multi-object detection on containing media data limitation. Amongst model capabilities, class-wise predictions and real-world waste management applicability are discussed. 3 Conclusion The chapter has demonstrated that the proposed system is able to produce reliable and scalable management waste for classification, detection performance.

Chapter 5

Engineering Standards and Design Challenges

In this chapter, engineering requirements, ethical principles and design obstacles involved in implementing the machine learning-based waste classification system are presented. It showcases how the project conforms to the well-known software/hardware/communication standards, which makes it reliable and ready for real-life deployment. The societal and environmental effects of the system and sustainability are also addressed in the chapter. Moreover, the project management details and challenging engineering problems encountered in implementation are presented to give a full understanding of the technical basis and ethical principle of the system.

5.1. Compliance with the Standards

5.1.1. Software Standards

The software part of WFDSS was developed according to standard engineering and coding practices so that it has been designed for long-term reliability, scalability, and maintainability. Python was chosen as the primary programming language due to support of machine learning, richnessf open-source libraries and clean-levelled syntax for structured programming. During the development phase, all implementation scripts were written in accordance with PEP 8 coding guidelines: readability, consistent spacing and indentation, clarity of naming conventions and standardization of formatting. Following them will guarantee that the whole codebase is in a good shape, easy to maintain and every future developer/researcher won't get lost in the spaghetti of your code.

In a bid to ensure that the model is decent and maintains good balance throughout, we have employed TensorFlow and Keras to construct the deep learning models. These frameworks run on a common computation graph, which is responsible for preserving the numerical stability even across hardware, so the behavior of model becomes reproducible when running code at different places. With using those standardized libraries, we could avoid system dependent errors and take advantage of the well-tested engine sights, autodiff and optimized gpu acceleration. This adherence to existing machine learning frameworks was impactful in making the model development cycle robust.

Pre-processing of the dataset and training of models was likewise organized based on reproducible engineering. These involved versioned pre-processing scripts with fixed directory structure, common labeling across all four waste classes. These procedures provided experimental repeatability, minimized the possibility of data mismanagement errors and allowed to trace back any changes in the project workflow up to their source. The YOLOv3 detection model used the standard Darknet annotation structure, making it compatible with most labeling tools and object detection workflows. No integration issues with existing frameworks were raised due to this adherence to annotation standards, and the detection subsystem was kept completely operational.

To enhance the flexibility and expandability of system, a modular structure software was taken. Every functional part (e.g., preprocessing, augmentation, training, evaluation and visualization) of the framework was implemented as an isolated module with a well-defined responsibility. This modular design concept helped us to update individual system components without the burden of altering overall system structure, a pre-requisite for long-term scalability. In addition, each stage was accompanied by detailed documentation which explained the modules, work flow diagrams and parameters used. This documentation promoted transparency, assisted debugging, and facilitated knowledge transfer when performing future upgrades or deploying the system in a real scenario.

5.1.2. Hardware Standards

The hardware specifications used in this system were selected for the support of the heavy computation requirements of deep learning–based waste classification and detection. A powerful computing environment was necessary due to big matrix operations, and a lot of parameters updates required in model training. Therefore, the development of the model and training was performed using the NVIDIA Tesla T4 GPU provided by Google Colab with 16 GB of VRAM for tensor operations. This GPU fulfills current industry criteria for deep learning research, and is computationally powerful enough to support training CNN models effectively as well as to keep numerical stability.

In addition to its GPU horsepower, the project made use of Google Colab's virtualized CPU and RAM infrastructure. The CPU allowed us to perform the image pre-processing (e.g., resizing, normalization and data augmentation) at a reasonable speed, while the system RAM capacity of 12 GB enabled us to load several thousands of images without worrying about memory overflow. These parameters enable compatibility with today's machine learning hardware, and produce workflow execution without any bottlenecks during training. The use of cloud delivered systems also reduced the problem of hardware dependency that is normally present in local machines.

In addition to the cloud services, hardware scalability was also considered in the system design. We chose lightweight architectures like MobileNetV2 and EfficientNet because we want to encourage running the models on devices with limited computational resources, even as simple as Raspberry Pi or NVIDIA Jetson Nano. These devices fallow strict hardware efficiency requirements including small form factor, low power and energy consumption as well as efficient on-device inference. By selecting models supporting them, we make the system ready for deployment in smart bins and recycling units.

Lastly, the hardware standards focused on reliability, interoperability and sustainability. All models and preprocessing utilities were verified to run on GPUs, CPUs, and embedded devices for risk mitigation with respect to hardware-dependent issues. Moreover, the architecture was cloud-native and could be easily replicated for experiments such that the entire system could be cloned and verified on different machines without any manual setup. This hardware compliance solidifies the system's robustness as a whole and makes it ready for academic research and deployment in practice.

5.1.3. Communication Standards

The exchange format provides communication standards for system components in the waste classification, to ensure their seamless interaction and data exchange without errors. The workflow below is modeled after common data formatting standards utilized by the community, so as to maintain consistency and interactivity throughout the pipeline. All datasets were saved as a series of images in common formats like JPG and PNG that provide light compression and are directly exploitable by machine learning frameworks. We maintained the labels and metadata in two widely readable, flat file formats with a lightweight structure, compatible with integrating them into data processing workflows from Python: CSV and JSON. Such consistency for file formats was a key reason that the training environment, preprocessing modules and evaluation scripts could interoperate smoothly.

For the object detection part, the Darket annotation style, which is the defacto standard for labelling YOLO models was used by the system. Each condition file provided class and normalized bounding box coordinates so that the model could learn object locations properly. Annotation using LabelImg guaranteed the compatibility with Darknet and prevented possible errors from label misalignment or wrong scaling. Using these common communication formats also facilitated the ability to swap, upgrade and retrain detection models in the future with minimal re-configuration.

For the sake of functional uniformity at model runtime, TensorFlow tensors acted as internal communication medium between preprocessing, training and inference parts. Tensors guarantees the uniform representation of data, providing stable computations across different CPUs and GPUs on the cloud. They also support batch processing for efficient execution, which is necessary for high-performance image classification and detection.

Lastly, documentation and reporting upheld the academic communication tradition to best communicate processes, results and system behavior. This encompassed structured logs, visualizations, and performance summary that allowed intuitive interpretation of model outputs. Combined, these communication standards provided an infrastructure for mature, portable and interactive collaboration among manifestations of the system at all levels of development.

5.2. Impact on Society, Environment and Sustainability

5.2.1. Impact on Life

This waste classification system is rather important to our life, which can be directly felt by us in terms of safety and hygiene. Simultaneously it also enhances the efficiency of trash management. Machine learning models are employed to automatically sort various types of waste, so that the health hazard generating from manual sorting will be decreased and cleaner living environment can be made possible. The mechanization not only safeguards waste workers from harmful exposure but also benefits the overall community with better sanitation. The key impacts include:

- Lower Worker Health Risks: The amount of time that people need to deal with hazardous waste directly is minimized by the automated identification process, preventing injury and exposure from harmful materials.
- Cleaner Living Environment: Classification can help improve the recycling effect, reduce the phenomenon of waste overflow, and beautify the city.
- Increased Productivity: Less of a need for workers to be trained and time required sifting through waste bins means waste management processes can flow more smoothly.
- Improved Public Hygiene: Appropriate waste management to control bad odors, preventing insects and disease (which benefits us).
- Elevated Life Qualities: Integrating waste incineration and disposal into urban construction, the system can improve the beauty of our cities by reducing environmental pollution; protect community health with these better solutions for waste treatment.

5.2.2. Impact on Society & Environment

The automatic system for sorting garbage significantly improves the lives of people and ecology, sorts waste better, recycled and processed. With machine learning providing accurate waste recognition the platform helps cleaner communities, and reduced pollution from mismanagement of waste. This promotes healthier living and responsibility for waste management in the society. The major impacts include:

- Improved recycling: Accurate sorting of recyclable materials (plastic, glass and metal) allows to identify proper separation and increase resource recovery.
- Minimal Pollution: Correct separation minimizes the amount of waste in landfills, contributing to les soil pollution, a reduced water table contamination along with less greenhouse gas accumulation.
- Tidier Public Spaces: By enabling waste authorities to better manage disposal, it makes for cleaner cities and increased sanitation.
- Supports for Sustainable Practices: The system upholds global sustainability objectives and environmental protection programs by supporting recycling, reduction of waste.
- Reduced Carbon Footprint: Efficient waste disposal results in low fuel use/energy consumption at recycling stations and less impact on the enviro environment of making virgin raw materials.

5.2.3. Ethical Aspects

The design of the classification model follows several ethical principles to ensure responsible use of artificial intelligence, fairness in handling of data and transparency in system operation. Because of the image-based datasets and automated decision-making used in the study, ethical considerations to ensure safety, privacy and reliability were is considered. With these measures in place, we can leverage the ongoing "successes" of our decision-supporting tools to help guarantee that a system is indeed doing more good than harm. Key ethical aspects include:

- Privacy-Compliant Data Utilization: All of the images used in this dataset are pure waste-contained objects — no privacy, or sensitive human data were captured or stored.

- Modelling with Fairness and Impartiality: The model did not get trained to discriminate waste categories and thus make biased predictions, which can harm the efficiency of recycling.

- Clear models: Your system should be so transparent you can follow a paper trail regarding which data was used during tearing and explain the thought process at inspection level.

- CONCLUSIONS Safe Use: The technology serves exclusively to improve the environment; it won't be used for anything bad, illegal or dangerous.

- Ethical Data Handling: Dataset curation adhered to a rigorous protocol in order to prevent mislabeling, redundancy or tampering of any sort in maintaining the reliability and relevancy of the model.

5.2.4. Sustainability Plan

The sustainability of the waste classification system's plan concentrates on the long-term technical reliability, operational continuity, and the burden to environment. Sustainability measures extend to cost and resources; the system entails machine learning models and a hybrid detection–classification flow, thus ensuring that the operational performance remains stable over the long-term while being as cost-effective and resource-efficient as possible.

Sustainability Outcomes:

- Technical Continuity:

  Frequent updates to the dataset and periodic retraining of the model contributes significantly in maintaining the accuracy of classification and YOLO model as they get new kind of waste images. The system is operating in a modular fashion, to allow the addition of new classes, better architectures or extra detection modules without repeating the entire pipeline.

- Operational Sustainability:

  Very little computation is needed by the system at runtime. Lightweight architectures such as MobileNetV2 or EfficientNetB0 successfully provide fast inference and they are suitable for long-term deployment on low budget hardware. There is some effort spent in documenting how things work, and to get them running using automated scripts that allow an easy maintenance of the environment because it can be used even by non-technical people.

- Environmental Sustainability:

  The project is also about promoting environmental consciousness by assisted to proper garbage disposal, reduce pollution, recycling. As effective classification aids in more efficient waste stream management, the system indirectly contributes to a cleaner environment and as such exhibits less stress on the landfill.

- Economic Sustainability:

  The system is cost-effective as it utilizes free tools (such as TensorFlow, Google Colab and open-source libraries). The system can be further scaled in future—both through integration at municipal recycling centers and into smart bins— with negligible investment, thus making the system economically feasible for sustained acceptance.

- Social Sustainability:

  The system motivates proper waste management behavior and promotes local awareness campaigns. It instills confidence and transparency in users, which encourages a continued adoption of eco-friendly habits.

5.3. Project Management and Financial Analysis

The project adopted a planned phase management approach that ensured smooth behavior and timely completion of tasks along with efficient utilization of resources. Each phase of the project, including data set preparation and processing, model build-ing and evaluation to documenta-tion would be baselined on an academ-ic calendar therefore having a clear milestone. At the level of individual tasks, workload was distributed and time managed depending on the nature complexity of the task. Economically, due to the adoption of open-source software, free cloud-based GPU resources (e.g., Google Colab) as well as either self-collected or publicly accessible datasets, this project was a prime example of being extremely cost-effective. The cost of internet and documentation was minimal, resulting in the software's overall financial stability and technical efficiency.

SN	Components	Estimated Cost (BDT)
01	Cloud GPU resources (Google Colab Pro/Equivalent for extended training sessions)	3,000 – 4,000

02	Local hardware maintenance (PC update, storage devices, accessories)	3,000 – 5,000
03	Internet and electricity costs during dataset preparation & model development	1,500 – 2,000
04	Software tools (paid licenses required; mostly open-source used)	1000 – 1,500
05	Documentation, printing, and binding	500 – 1,000
06	Contingency (10% of total cost)	900 – 1,350
	Total Estimated Cost	9,900 – 14,850 BDT

Table 5.1: Actual Academic Budget.

Alternate Budget: Hypothetical Deployment

The operation and infrastructure cost that would be incurred if the waste classification system was to be implemented in real-world setting, including running it under Dhaka North City Corporation (DNCC), or Dhaka South City Corporation (DSCC), as well as smart recycling plant. The estimated budget parameters are listed in the following table e for a theoretical one-year operation of the new automated waste classification and detection system.

Table 5.2: Alternate Budget.

SN	Components	Estimated Cost (BDT)
01	Edge hardware devices (e.g., Raspberry Pi / Jetson Nano for smart bins)	25,000 – 40,000
02	High-resolution cameras for continuous waste monitoring	15,000 – 25,000
03	Cloud/Server hosting for running ML models & data storage	12,000 – 20,000 / year
04	Maintenance, repairs, and system updates	8,000 – 12,000
05	Internet connectivity & operational electricity cost	6,000 – 10,000
06	Staff training & technical support	5,000 – 8,000
	Total Estimated Cost	**71,000 – 115,000 BDT**

Table 5.2: Alternate Budget for one year.

5.4. Complex Engineering Problem

5.4.1. Complex Problem Solving

We developed a systematic solution to the complex problem of nostril detection that used state-of-the-art machine learning techniques and some experimentation. Data variability and class imbalance were addressed by pre-processing, augmentation, and proper dataset arrangement. Various models were trained (MobileNetV2, ResNet50, EfficientNet and YOLOv3) to find the optimal model for classification as well as detection. Iterative refinement, performance evaluation, and hyperparallel tuning were utilized to achieve maximum accuracy and robustness. Through collaboratively connecting multiple modules, which are interrelated with each other, into an integrative pipeline, this system has effectively addressed the complicated automation engineering problem of waste identification.

EP1 Depth of Knowledge	EP2 Range of Conflicting Requirements	EP3 Depth of Analysis	EP4 Familiarity of Issues	EP5 Extent of Applicable Codes	EP6 Extent of Stakeholder Involvement	EP7 Interdependance
√	√	√	√		√	√

Table 5.3: Mapping with complex problem solving.

Justification for EP Attributes Mapping

- EP1 – Depth of Knowledge: The work needs to have a strong working knowledge of deep learning, CNN architectures, transfer learning and YOLO based object detection.
- EP2 – Conflicting Requirements: The system must be able to maintain the accuracy-trade-off-speed, low-cost model-high performance model trade-offs, and process data imbalance in different waste categories.
- EP3 – Depth of Analysis: Several models were trained and evaluated with accuracy, precision, recall, F1-score, mAP, confusion matrix because of informed analytic judgement.
- EP4 - Ease of Issues (Pertinent in Parts): Image classification issues are less of a concern since waste materials come in new types, e.g. transparent, overlapped ones and problematic lighting conditions.
- EP5 – Applicable Codes (Limited): There is no formal set of regulatory codes for waste identification but the project follows coding standards (PEP8), YOLO annotation formats, and best practices for ML.

- EP6 – Stakeholder Involvement: The service life and utility of the combined system are affected by stakeholders (i.e., collectors, recyclers, and local governments) with practical requirements.
- EP7 – Interdependence: The project has multiple dependencies preprocessing, augmentation, classification and detection which together function as one unit.

Mapping with Knowledge Profile for EP1

This table 5.4 is designed to map the EP1 to the Knowledge Profile.

K3 Engineering Fundamentals	K4 Specialist Knowledge	K5 Engineering Design	K6 Engineering Practice	K7 Research Literature
√	√	√	√	√

Table 5.4: Mapping with knowledge Profile for EP1.

Justification for Knowledge Profile Mapping

- K3: Engineering Fundamentals: Includes good fundamental backgrounds in python programming, image processing, CNN and evaluation metrics for classification/detection.
- K4: Specialist Knowledge: Advanced knowledge of deep learning model architecture such as MobilenetV2, ResNet50, Efficientnet and YOLOv3 is essential along with model optimization, real-time waste detection experience.
- K5: Engineering design: Designing of a full machine learning pipeline, from preprocessing image data and to classification or detection to integration in the system for practical waste-classification applications.
- K6: Engineering Practice: Including the software execution flow of data collection, annotation, model training on GPU platforms as well as realistic variations like lighting changes, clutter and class imbalance.
- K7: Literature Research: Extends the existing research on waste classification and object detection by using proven methods to produce a working real-world solution, rather than proving new theory.

5.4.2 Engineering Activities

The development of the AI-based waste classification system required various types of engineering work, such as data treatment, model design, system architecture integration, environmental aspects and technical decision making. All these activities have been essential to get a final system which is efficient, reliable and usable in real scenarios. The reasoning for this mapping from engineering activities to the project is detailed as follows.

Table 5.7: Mapping with engineering activities.

EA1 Range of Resources	EA2 Level of Interaction	EA3 Innovation	EA4 Consequences for Society and Environment	EA5 Familiarity
√	√	√	√	√

Rationale for Mapping Engineering Activities

- EA1: Range of Resources: The work was implemented using different open-source tools, including TensorFlow, OpenCV, Keras and Albumentations libraries as well as a Google Colab GPU platform. These tools facilitated data preprocessing, augmentation, classification and object detection in a computationally effective manner.

- EA2: Level of Interaction: The system is intended for use by waste pickers, recycling plants and municipal bodies where a simple and intuitive interface are needed. Image uploading Data processing facilitation, object detection visualization and classifying results are available for an effective interaction and daily use.

- EA3: Innovation: It forms a hybrid classification-detection pipeline using several deep learning models MobileNetV2, ResNet50, EfficientNet and YOLOv3. This novel fusion can perform both object level and real-time multi-object waste detection.

- EA4: Implications for Society and Environment: It not only makes handling of garbage safer, but is also more efficient in recycling waste and reduces pollution by lowering the incidence of misclassification. But the impact it has, clean cities, improved public hygiene or long-term sustainable projects.

- EA5: Familiarity: Although inspired by existing machine learning architectures, the framework needed to be customized to address local types of waste, lighting conditions, background clutter and data sparsity: these elements make it well suited for real-world waste management problems in Bangladesh.

5.5 Summary

The engineering criteria, social implications & design issues for development of the waste classification system were illustrated in this chapter. It showed how a system of software, hardware and communications specifications could be followed for reliability, interoperability in both the short and long term. The social and environmental consequences of the project were also addressed, with particular emphasis on better waste management, diminished pollution, and more secure working conditions. Furthermore, the chapter gave attention to how the system was engineered and mapped this to suitable knowledge profiles and engineering activities. Together, these arguments in common affirm that the system is a responsible, technically rigorous and socially desirable engineering response.

Chapter 6

Conclusion

This paper concludes the summary of the waste classification system using machine and deep learning. It gives an overview of the accomplishments, main results, and deliverables of the project and discusses how well the executed models performed. The chapter ends with a discussion of the drawbacks made during its construction, and by suggesting some improvements. Together, they provide a full picture of what the project achieved, and its readiness for deployment in sustainability practices regarding waste management.

6.1. Summary

The process of machine learning–based waste classification system originated by painstakingly gathering and preprocessing a random database that holds more than 3000 images in four different classes of waste: Plastic, Metal, Glass, and Polithin. A combination of self-captured images and publicly available sources were used to create the dataset in order to have a diversity of lighting, orientation, background and waste appearance. Each image was subjected to preprocessing including resizing, normalization, augmentation etc., in order to make the model more robust and versatile. This well-organized dataset served as the training of both classifiers and detectors in the project working.

3.1 Training phase First a hybrid architecture was defined, by adopting three transfer learning–based classification models (i.e., MobileNetV2, ResNet50 and EfficientNetB0/B1) and fusing them with the YOLOv3 object detection framework to form the implementation stage. Although the models used for classification were developed to find a single waste item, added YOLOv3 was used for object detection of multiple items in real-time with bounding boxes as outputs. All models were trained in Google Colab's GPU support environment, fast computation under stable convergence. Advocated such practices of standard coding, modular and reproducible experiments setup throughout the process to ensure system reliability/ scalability.

The results of model performance on the test set showed that each model had great classification abilities, with some significant differences in precision in overall accuracy. MobileNetV2 obtained a testing accuracy of 84.72%, indicating its light weight and speedier inference. EfficientNetB0 also led to a better performance of 86.32%, whereas resnet50 considerably improved this value to 88.95%. Of all the architectures tested, the accuracy of EfficientNetB1 was greatest: 89.12%, indicating that this model had better feature extraction ability and generalization performance. Meanwhile, YOLOv3 achieved a mAP@0. 5 out of 62.40% thereby accurately detecting object multiple instances even in dataset challenges, either background noise or class similarity.

The overall results have proven the effectiveness of classification models in the case of single waste images, while with YOLOv3 it becomes easy for our system to operate successfully on mixed-waste categories existing in real-world scenes. The incorporation of these models indicates the potential in automating waste-sorting activities and mitigates the unhygienic and unproductive nature of manual handling of trash. And it all makes good sense, in terms of achieving sustainable waste management: increased recycling efficiency, employee safety improvements and the development of smart city programs that value environmental protection.

In the end, the project was able to achieve its goals by developing a fully functioning system that was designed, implemented, and evaluated using state of the art deep learning techniques. Comparison between the models offer insights on their strengths and weaknesses that could facilitate the selection of appropriate architectures in future waste management applications. Its hybrid architecture together with its promising accuracy demonstrate the potential of the system to be used in smart bins, sorting facilities and other environmental sustainability infrastructures.

6.2. Limitation

Despite the positive results obtained, several limitations are associated with the waste classification system in terms of performance and application on industrial level. One large scale limitation of the dataset project is size and diversity of the dataset. Since database is large, with more than 3000 images, we had fewer high quality samples for some classes – especially Polithin and that lead to mis-classifications and less recall of the said classes. Due to lighting and background variation, or the translucency of items, some images of waste were also visually ambiguous and did not allow for easy identification by our models.

Another drawback of the models is related to their characteristics. Despite their accuracy, EfficientNetB1 and ResNet50 remain computationally intensive as they cannot yet be deployed in extremely low-resource settings. In contrast, MobileNetV2 is a lightweight network but it loses some accuracy and is not desirable in precision-required applications. Although the YOLOv3 was successful for detection, mAP performance is moderate because the training data is small and it cannot detect partially hidden or overlapping waste objects.

Environmental variation presents another challenge. Real waste scenes involve unexpected factors such as mixed waste types, shadows, occlusion, dirt or background clutter that were not accurately shown in the controlled training environment. Then the performance of such models may fall when they are deployed at uncontrolled or cluttered scenes. Moreover, the manual annotation of YOLO detection was time-consuming and error-prone to make it an impact on the performance of detection.

Last, the system's capabilities are limited to only four different waste types, limiting its applicability in a more general recycling environment with multiple dozens of waste varieties. Introducing more classes would be achievable only after significant dataset addition, accurate labeling and further training of the model. Among the limitations of the project, there was no citation to image processing, deep learning algorithm or intelligent system that can be classified as relatively accurate and reliable design for waste classification.

6.3. Future Work

- Extend the dataset so that it includes a wider variety of waste categories and realistic conditions like changing illuminations, backgrounds and scenarios with mixed waste types while ensuring sufficient generalization of the model.
- Improve the detection model with state-of-the-art architectures like yolo v5 or yolo v8, for better accuracy and to handle overlapping objects.
- To make a real-time deployment design, attempt to link the system with some smart garbage bins, or embedded devices such as Raspberry Pi or Jetson Nano for on-site waste sorting.
- Adopting segmentation methods, such as Mask R-CNN, for identifying waste at pixel level to ensure accurate classification and improve the recycling efficiency.

This chapter finalized the prototype of the waste classification system by reflecting on its main contributions, describing current limitations and suggesting future developments.

The work successfully showed the power of deep learning for automatic waste sorting and suggested dimensions that need to be improved such as data enrichment, sophisticated detection models and on-line deployment. In general, the system provides a promising platform for future research and practical application in sustainable waste management.

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